

Materials for Critical Imagination

Grace Uribe

Interaction Design
One-year master
15 credits
Semester 2 / 2024
Supervisor: Per-Anders Hillgren

Abstract

This study explores how Interaction Design could contribute to facilitating imagination and critical thinking in participatory settings. Specifically, it investigates the intersection between Playful Design and Design Critical Practices as a way to support the Re-Imaginatorium, which is an initiative led by Malmö University that aims to facilitate participatory spaces for Critical Imagination. Consequently, the Re-imaginatorium is the main stakeholder and the contextual frame of this study. From the theoretical analysis, the notion of ambiguity was found as a bridging concept between Playful Design and Design Critical Practices for supporting imagination. This was a relevant tool for developing different interactive materials for three Re-Imaginatorium pilots. Different prototypes of materials were created for each pilot. The last one incorporated multisensorial elements, including visual, sound, smell, and tactile resources. This last one demonstrated that the concept of ambiguity combined with interactive and multisensorial resources, enables a playful attitude, that in turn could potentially contribute to critical and collaborative imagination. Feedback from each test was incorporated into a concept of a ToolKit for the Re-imaginatorium.

Table of contents

1. INTRODUCTION **4**

1.1 Aim and Research Questions 5

1.2 GDPR Disclaimer 5

2. CONTEXT **5**

3. THEORY **6**

3.1 Play and Playfulness 6

3.1.1 Playful Experience 7

3.2 Appropriation & Ambiguity for Critical Imagination 7

3.2.1 A taxonomy of ambiguity in design 8

3.3 Materials for Critical Imagination 8

3.3.1 The play Object 9

3.4 Game Materials in Participatory Settings 9

3.4.1 Critical Play – Radical Game 9

3.5 Canonical Examples 10

4. METHOD **15**

4.1 Project Plan 15

4.2 Design Process Model 16

4.3 Discovery 17

4.4 Design 18

4.5 Implement 19

5. DESIGN PROCESS **20**

5.1 Discovery 21

5.2 Design 23

5.3 Implement 25

6. MAIN RESULTS AND FINAL DESIGN CONCEPT **42**

6.1 Main Results from the Design Process 42

6.2 Design Concept 42

6.3 Answers to Research Questions 45

6.4 Knowledge Contribution 47

7. EVALUATION **47**

8. CONCLUSION **48**

1. INTRODUCTION:

Imagination sparks change by letting us dream beyond what is currently possible to envision alternative realities (Moore & Milkoreit, 2020). It supports the complex processes of tackling major challenges and enhances the capacity of individuals and communities to create, shape, and inform decision-making across social-ecological dimensions (Moore & Milkoreit, 2020). Along with the emphasized importance of imagination for our current societies, various scholars have pointed out that our capacity to imagine alternative ways of living is facing a significant crisis (Unger, 1987; Harvey, 2000; Kiersy, 2013, as cited in Lindström et al., 2021). Consequently, there is a need to promote and cultivate a type of imagination that allows us to envision not only alternative but also sustainable and equitable futures (Lindström et al., 2021).

One initiative that looks to cultivate this type of imagination is the Re-Imaginatorium. It is a format led by Malmö University and is currently in the development phase. The Re-Imaginatorium aims to open collaborative spaces for, what the creators called, Critical Imagination, a way to challenge taken-for-granted assumptions and worldviews, facilitating the emergence of critical questions. The Re-Imaginatorium serves as the contextual framework in which this thesis project is conducted (See section 2).

However, the challenge for scholars and practitioners is to reveal how imagination is cultivated and what factors sustain it. Different discussions have recognized that even when in general terms imagination is associated with freedom, it simultaneously faces various ways in which its freedom is constrained (Kind, 2022). Some limits come from our nature as humans and some others from the lack of experiential resources or an untrained skill (Kind, 2022). Aligned with Kind´s arguments other authors highlight the essential role that materiality, social settings, and embodiment have in supporting imagination (Smith et al., 2021)

Interaction Design practitioners have supported these three aspects by creating tangible artifacts that through embodiment allow personal and shared interactive experiences. Accordingly, this study aims to explore how Interaction Design can support the pursuits of the Re-Imaginatorioum.

The Re-Imaginatorium had its first pilot in December 2023, it was a one-day-long workshop where different actors from the city (Public sector, Academy, and Non-profit organizations, among others) came together to re-imagine Malmö in 2040. This experience brought valuable insights into diverse aspects of the format. On one hand, engagement and participation were core and successful aspects. On the other hand, it was reaffirmed that it is challenging to move participants out of the current social and mental structures to imagine how things could be otherwise.

Some authors have reflected on the possibilities that Playful Design can bring into this regard (Skovbjerg et al., 2024). Applying qualities from play into different designs can help to create opportunities to meet the unknown future with curiosity, expression, and social interaction (Skovbjerg et al., 2024). For example, Sicart et al. (2014) see play as a way to bring occasional freedom and distance from conventional social structures. Furthermore, through playful appropriation, this freedom can be allowed in different contexts where play is not the only purpose (Sicart et al., 2014), like the context of the Re-Imaginatorium looking to promote critical reflection.

The combination of these possibilities and the various elements of the Re-Imaginatorium inspired this thesis project, forming the foundation for the research objectives, questions, and contributions outlined below.

1.1 Aim and Research Questions

This thesis project has the aim to support the process of creating the format of the Re-imaginatorioum and its materials. This is by examining how playful characteristics can be applied in the design of materials that enable Critical Imagination.

Research questions:

1. What role could Playful Design have concerning speculative thinking in a participatory setting?

2. How Playful Design can make sense in the critical reflection and future creation of different aspects of human ways of living?

3. What is the relation between playfulness in critical design practices?

1.2 GDPR Disclaimer

This research has been conducted in line with GDPR legal stipulations. Any other private information that can lead to the discovery of the identity of the persons involved in the user testing workshops has been revealed.

2. CONTEXT

Malmö University is part of the UNIC - The European University of Cities in Post-Industrial Transition. “This is an alliance of ten universities educating through teaching, research and community engagement, towards inclusive societies” (UNIC, 2024). Its mission is to foster the development of knowledge and co-creation through collaboration, innovation, and inclusion, to bring about positive societal impact and improvement. UNIC has the vision to take inclusion at the core of the alliance by offering educational opportunities for every student regardless of their background while aiming to acknowledge and break the privilege structures that constrain access to several students. Additionally, it engages students in real-world urban challenges through initiatives such as City Labs. These are physical and virtual spaces, where students, academia, citizens, and other city stakeholders can collaboratively identify and solve societal challenges.

The Centre for City Futures (CFF) is an important element within UNIC and aims to be the platform that supports co-creation and close collaboration between the universities, the cities, and citizens in the ten UNIC cities. It also brings together the capacities and methods to co-create, like the mentioned City Labs.

As part of this alliance and within the CFF, Malmö University is leading the development of the Re-Imaginatorium as a way to support the co-creation efforts of UNIC. As mentioned in the introduction, the Re-Imaginatorium is a format that looks to open up participatory spaces for critical imagination. In different sets of encounters and through a collaborative approach, the Re-Imaginatorium invites researchers, teachers, students, citizens, and other actors to re-imagine the world and its diverse everyday aspects. Often, these re-imaginations take the form of scenarios and start with radical “what if” questions.

This format is in the process of development and had its first pilot in December 2023. Throughout the time frame of this thesis work, it hosted three other events in May 2024.

3. THEORY

This research uses play and playfulness concepts, characteristics, and techniques as the backbone for exploring new ways of supporting the dialogue of critical questions and the re-imagination of future scenarios. In this section, those concepts are introduced and linked to the Critical Design practices by their common use of the concept of ambiguity. Furthermore, it introduces the use of materiality for promoting imagination in participatory settings. Finally, it ends by presenting three examples, one about the use of ambiguity in Speculative Design, the second related to the use of materials in participatory settings, and the third illustrating the use of materials for imagination.

3. 1 Play and playfulness

A considerable amount of theorization of play has been derived from the concept of Homo Ludens by the work of cultural historian Johan Huizinga, from this view play is a fair contest that creates a world with its own rules and is separate from real-life (Sicart et al., 2014). Although his work still influences newer approaches to play, this thesis project draws from the view proposed by Miguel Sicart as it integrates play into human everyday reality and its different contexts. Under this line, Lucero & Arrasvuori (2010) also state that most human activities, even pragmatic or mundane tasks, can be approached and experienced to some extent as a form of play.

Sicart et al. (2014) responded to the instrumentalized and mechanistic approach to play in the postmodern culture industries by conceptualizing play as "a mode of being human" (p. 1), "as a way of expression and engaging with the world— not as an activity of consumption but as an activity of production" (p.5). As mentioned in the introduction, he considers that humans need occasional freedom to step back from our conventional moral structures, and he sees play as a way to do so. Furthermore, Sicart et al. (2014) argue that play allows us to go against efficiency, seriousness, and technical determinism, aptitude needed when creating new futures or promoting change. Aligned with this creative approach, other authors have stated that play has been recognized as a way to achieve innovation and creativity because it enables people to see things differently or make unexpected connections (Fullerton et al., 2004).

On the other hand, Sicart et al. (2014) define playfulness as "the capacity to use play outside the context of play" (p. 21). For them, playfulness is a way of engaging with particular contexts and objects that is similar to play and has some of its characteristics, but, it does not change the purposes and goals of that object or context. "Playfulness is a physical, psychological, and emotional attitude toward things, people, and situations" (Sicart et al., 2014, p. 21).

These two concepts overlap, but they do not refer to the same thing. The main difference between play and playfulness is that play is an activity, while playfulness is an attitude (Sicart et al., 2014). An activity is a coherent and limited group of actions performed for certain purposes (for example, to play with the purpose of playing), while an attitude is a stance toward an activity, the personal viewpoint from which we approach activities, people, and objects (Sicart et al., 2014). Similarly, Fullerton et al. (2004) state that a playful approach can be used even in the most difficult or serious contexts because "playfulness is a state of mind rather than an action" (p.88). In this idea, playfulness means adding characteristics of play to nonplay activities (Sicart et al., 2014). "Playfulness means taking over a world to see it through the lens of play, to make it shake and laugh and crack because we play with it" (Sicart, 2014, p. 24).

3.1.1 Playful Experience

Korhonen et al. (2009) argue that "playful experiences emerge from interactive products that allow users to have a playful approach while using them" (p. 277). Furthermore, Lucero & Arrasvuori (2010) state that obtaining this type of experience may per se be highly motivating. Additionally, Korhonen et al. (2009) argue that to comprehend playful experiences it is important to study the notion of enjoyment. Consequentially, it is important to understand what aspects constitute the enjoyment of using those interactive products, what kinds of experiences they can elicit, and how to design products with properties that promote that kind of experience. Based on this, Korhonen et al. (2009) have developed a Playful Experience Framework, taking as reference the Pleasurable Experience Framework created by Costello and Edmonds (2007). The reason behind this decision was Costello and Edmonds´s approach of going beyond the usual user experience paradigm that only emphasizes usability issues (Costello & Edmonds, 2007). Aligned with this, Korhonen et al. (2009) have expanded this approach by reviewing studies of relevant game experiences and merging all of them into 19 playful experience categories (See Appendix 1.3). These categories can be used in the moment of designing products that elicit playful experiences, and respond to the analysis and discussion of why people play and what the properties of playful products are that attract people into using them.

3.2 Appropriation & Ambiguity for Critical Imagination

One core characteristic that play and playfulness have in common is appropriation. Sicart et al. (2014) explain that by playing we take over a context, we appropriate it in a way that it becomes "the servant to the activity of play" (p. 11). Furthermore, for the activity of play to exist it not only appropriates the context but the objects it uses. Likewise, he explains that to be playful is also "to appropriate a context, that is not created or intended for play. Playfulness is the play like appropriation of what should not be play" (p. 27).

Appropriation implies a shift in the way we interpret a particular situation, technology, or object. Appropriation is enhanced when the context and objects enable open or multiple interpretations (Sicart et al., 2014). One characteristic that supports these types of interpretations is ambiguity. Sicart et al. (2014) argue that playfulness reambiguates the world. For him, "to be playful is to add ambiguity to the world and play with that ambiguity" (p. 28). However, even when the potential of the aesthetic and conceptual aspects of ambiguity has been long used in arts, the traditions in design have maintained efforts to prevent playfulness, in Sicart's words "resisting by design the temptation of appropriation" (p. 30) and therefore it is seldom embraced within traditional Human-Computer Interaction (HCI) (Gaver et al., 2003).

Even when ambiguity could be seen as the opposite of usefulness and usability, the two goals of HCI research and development (Nickerson & Landauer, n.d., as cited in Gaver et al., 2003), it has been proved that it can be a valuable resource in the design of interactive products that are both engaging and thought-provoking (Mathew & Taylor, 2008). Gaver et al. (2003) agree that it is not easy to see how a product is usable if the information it presents is unclear, or how it is useful if its purpose is uncertain, however, they understand ambiguity not as a problem but as an opportunity to promote curiosity, exploration, and reflection.

Allowing this ambiguity to be incorporated into the design of a product or system has several advantages (Gaver et al., 2003). It allows designers to express their points of view and at the same time enables users from diverse socio-cultural backgrounds to find their own interpretations. By promoting users to generate their interpretations a deeper and more personal relation with the meanings offered by those systems is promoted (Gaver et al., 2003). "The most important benefit of ambiguity, however, is the ability it gives designers to

suggest issues and perspectives for consideration without imposing solutions" (Gaver et al., 2003, p. 240). In this way, objects and systems become a resource for people to appropriate (to complete) and make them meaningful, rather than content for consumption or a tool for the performance of a defined task (Gaver et al., 2004).

3.2.1 A taxonomy of ambiguity in design

Critical Imagination has its roots in Speculative and Critical Design. The following taxonomy of ambiguity developed by Gaver et al. (2003) and expanded by Malpass (2013) serves as a reference for this thesis project as it shares those roots and incorporates Critical Design practices (Speculative, Associative, and Critical Design) into three kinds of ambiguity in design.

Each kind of ambiguity asks for different sorts of interpretation. 1. Ambiguity of information: This type of ambiguity asks users to interpret by filling in the gaps of incomplete information. It makes people question the truth of a situation (Gaver et al., 2003). Malpass (2013) argues that Speculative design works through this type of ambiguity as it portrays new theoretical and applied technologies that are often unfamiliar. This work relies on creating external narratives and scenarios that illustrate the potential uses of a design.

2. Ambiguity of context: It requires an integration of what could be seen as incompatible frames of reference. "Contextual ambiguity can question the discourses surrounding technological genres, allowing people to expand, bridge, or reject them as they see fit" (Gaver et al., 2003, p.240). Malpass (2013) states that Associative Design works through this type of ambiguity as it works through the subversion of objects or their context and ways of use.

3. Ambiguity of relationship: This last type, evokes an interpretation from subjective experiences into new situations. It can lead people to consider new beliefs and values and their own. This type of ambiguity in contrast to the other arises from the viewer's personal relationship with the object or piece (Gaver et al., 2003). Malpass (2013) relates it to Critical Design as for this type of design users need to experience a kind of dilemma and carry something of a burden of interpretation. Here objects seem to be real but something feels not quite right. These mechanisms prompt the question of what would be needed to change a current reality.

In each of these kinds, ambiguity allows users to freely react to designs with skepticism or belief, appropriating them into their own lives through their interpretations (Gaver et al., 2003). In the process of reacting to the designs, either positively or negatively, users engage with matters that the designer suggests. This reveals that ambiguity is a powerful design tool for bringing up to the surface challenging topics or asking questions while being against dictating fixed answers. By this balance, ambiguity offers an inspiring resource to designers while demonstrating respect for users (Gaver et al., 2003).

3.3 Materials for Critical Imagination

As stated by Kind (2022) experiential resources can facilitate the imaginative process, furthermore as just described, using ambiguity as a tool when designing objects and systems can promote personal interpretation while enhancing critical reflection (Gaver et al., 2003; Malpass, 2013). Smith et al., (2021) also aligned with the idea of bringing tangibility to scaffold imagination and expand it by highlighting that it should be embedded in an embodied and social experience.

3.3.1 The play object

According to Sicart et al. (2014), a play object is a prop for a play, it can be a game or a toy. These types of objects as many others have a purpose and a function, however, they achieve this through the personal and appropriative act of playfulness. “They do not become toys or pure playthings, but the behavior and attitudes toward them, the ways they redefine the contexts in which they are applied, invoke the characteristics of play” (Sicart et al., 2014, p. 32). Playful objects are designed for appropriation as they enable openings in a particular context or situation that justifies the activity of play (Sicart et al., 2014).

Toys are objects that embody the freedoms of play (Sicart et al., 2014). As playful designs, toys maximize their potential when they are ambiguous and open to multiple interpretations. In this way, they are like empty vessels to be filled with stories, worlds, and actions that come from the imagination of who is using them (Sicart et al., 2014).

Toys suggest or bring hints to users about how to interact with them, but they never enforce a behaviour. Differently happens with games which leads to a more formalized play (Sicart et al., 2014). Anyhow, games also host appropriation qualities. It promotes appropriation through tactics. When players in a game create tactics, those could be part of the game or personal, they make interpretations of that context and their ways of interaction within it. “The tactics are the on-the-fly creative interpretation of a game through the activity of playing it” (Sicart, 2014, p. 17).

Moreover, the terms “play” and “game” refer to two intertwined, but different things. To differentiate them, some scholars have used the concepts of “paidia” and “ludus” which refer to free and formal play, accordingly (Korhonen et al., 2009). Paidia and Ludus can be seen as the opposite ends of a range of activities. Ludus is playing with a set of rules that determine who wins and who loses, while paidia is a type of play that is not limited by a fixed set of rules and winners and losers (Frasca, 2003, as cited in Lucero & Arrasvuori, 2010).

3.4 Game Materials in Participatory Settings

Games have been proven to facilitate collaboration in design processes with diverse purposes (Brandt & Messeter, 2004). In particular, Exploratory Games, help organize participation in settings involving people with diverse backgrounds, competencies, and professional language (Brandt, 2006). They are engaging and create an informal atmosphere which is known to foster creative work. They enable the stakeholder's various competencies to be utilized so everyone can make design contributions and be part of exploring and negotiating views to create common images of possible futures (Brandt, 2006). In this setting, the game materials create a common ground to which the different stakeholders can relate to, therefore they become “things-to-think with” (Papert, 1980, as cited in Brandt & Messeter, 2004) where the reflections from different participants result in re-seeing the task, to give it a new meaning. Furthermore, these game materials can be understood as boundary objects (Star, 1989, as cited in Brandt & Messeter, 2004) that share an essence that participants can easily relate to and also are open enough to allow different interpretations.

3.4.1 Critical Play - Radical Games

Even when many game players consider games as mediums for entertainment, fun, or just to pass the time, the more general understanding of play expands the contexts and purposes of it. Critical play - Radical games take the general characteristics of play to develop games that function as means for creative expression, acting as instruments for conceptual thinking or as

tools to support discussion and work around critical matters (Flanagan, 2009). “Critical play means to create or occupy play environments and activities that represent one or more questions about aspects of human life” (Flanagan, 2009, p. 6).

Materials and dynamics in this kind of game provide an essential viewpoint or an analytical framework, related to the select issues or subjects, to promote critical thinking. As with any other type of game it enables decision-making and agency, in this case serving the values the game designer has selected to touch upon. Defining the values and the rules or actions around them, are crucial as they are the two main characteristics that distinguish radical games from other types of games (Flanagan, 2009).

3.5 Canonical examples

Three main examples have been selected concerning the main theoretical concepts. One shows the use of ambiguity for thought-provoking artifacts within the field of Speculative Design. Another explains the use of materials for setting up participation. The last one describes the use of materials to support imagination.

3.5.1 Ambiguity Used in Speculative Design

Robot 1 from the Technological Dreams Series: no 1, Robots project (Model). 2007. By Anthony Dunne and Fiona Raby.



Figure 1. Technological Dreams Serie by Anthony Dunne and Fiona Raby.

This is a series of four strange-looking objects, which are called robots. However, as they do not look like as the archetypical idea of a robot, people do not look at them as if they were actually robots. These are thought-provoking objects that spark questions and reflection on people about how robots should look in the future, which technologies should they represent or manifest, and how they might affect the lives and bodies of people. They also make people wonder "Um, why can't robots look like this?" (Dunne & Raby, 2007). These types of imaginary design products look to promote debate and discuss the future possibilities of new technologies

and the emotional relationship products at home can have with their users. Furthermore, to speculate and explore what people are attracted to or repelled by (Dunne & Raby, 2007).

Based on Malpass´ (2013) arguments about the Ambiguity of Information, it can be argued that this example represents the basics of this type of ambiguity. These objects whose formal manifestation and use are not evident enhance people to make interpretations to fill in the information not present or clear. For example, information about the purpose, the ways of use, the functionality, and even about its real existence. Furthermore, these kinds of objects are also inspirational and support overcoming the division between the "real real" and the "unreal real" unlocking people´s imagination about how our world could be (Dunne & Raby, 2013). This resonates with the idea and roots of the Re-Imaginatorium.

3.5.2 Game materials in Participatory Design projects

Exploratory games by Brandt and Messeter (2004)

From the action-based research about how to successfully involve diverse stakeholders in a participatory design process at Space Studio, a set of games has been developed and put into practice in different contexts (Brandt & Messeter, 2004). This set consists of four games that are suggested to be played in a specific order as they build on each other.

1. The User game: In this game, the players should create stories about prospective users of a new technology. These stories are grounded in field video data collected from ethnographically inspired field studies.

- Game materials: Two types of game pieces called 'Moment-cards' and 'Sign-cards' (Figure 2). The Moment-cards are numbered, with each one corresponding to a video clip from the field data. The Sign-cards are used to label the stories created. They have a word printed, for example: 'despair', 'pace', 'vibrant', 'closeness', and 'zones'. The purpose of the Sign-cards is to provide a conceptual framework for the stories. Therefore, they change depending on the project.

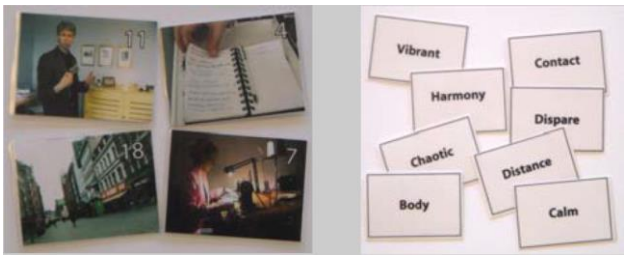


Figure 2. 'Moment-cards' and 'Sign-cards'

- How to play: The first player receives an amount of Moment-cards and creates a story based on those. The following player chooses two to four Moment-cards and creates a second story and one Sign-card as a label for the story. This second story is added to the board game by placing the sequence of cards in a way that crosses the first one. The card at the intersection must be part of the story. This is repeated by each player forming a crossword-like structure on the game board (Figure 3).



Figure 3. Participants looking at the videos.
Crossword-like structure on the game board.

2. The Landscape Game: Here players create a context for the individuals described in the first game by highlighting the physical surroundings or elements that enhance various activities in the person's daily life.

- Game Materials: They include Game-boards with different generic conceptual maps. Traces cards that have pictures of the physical surroundings from the field material, and the same Moments cards used in the User Game. (Figure 4).



Figure 4. Game-boards. Moments cards.

- How to play: The players choose one of the game-boards to work with based on mutual agreement. Then they select two users who have different ways of working and place them in a specific part of the game-board as they see fit better. After, players watch the video clips corresponding to the Moment-cards they chose. They discuss and place on the game-board too. This process continues by discussing the rest of game-pieces and placing them in communal agreement on a certain space on the game-board (Figure 5).



Figure 5. Participants looking at the videos.
Moments cards on the game-boards.

3. The Technology Game: In this game technologies and the possible form of ideas are introduced and investigated. Here players create scenarios about the people in the previous games involving functionalities and assigning or creating a shape that represents the possible product. The description of the scenario is written on paper.

- Game Materials: The game pieces can vary; however, they should illustrate technology and form aspects. As an example: Foam/cardboard that can be easily shaped and LEGO pieces.

- How to play: The players discuss and agree on how to formulate the scenario and write it down. Then, by using the foam models, cardboard, and/or the LEGO-Duplos generate, explain, and argue about ideas or potential design concepts.



Figure 6. Different game materials.

4. The Scenario Game: In this last game, the experiences from the previous games are synthesized into scenarios involving people, context, activities, and the technology to be designed. These scenarios are built and performed on-site. The first three games take place on a table. It has the purpose of nurturing empathy for the users and the situations of use and being able to adjust the design concepts while acting out scenarios in situ.

- Game materials: They are most often simple versions of the artifacts proposed in the design concepts. They are usually created in simple cardboard or foam forms.

- How to play/enact: The players get a set of generic props and then in agreement after reflection, they select the one they want to use to enact the scenario in situ. Players are also encouraged to create their own versions of artifacts.

This example was selected as it has long been recognized as a reference for the creation of games and materials when setting design participatory processes. This approach has also been used several times and in very diverse contexts with stakeholders from various backgrounds, from the private and public sectors. Furthermore, in these kinds of games, materiality is a core aspect. The game material is designed to be ambiguous in most cases allowing openness for various views and understandings. They are intentionally not referring to a real-life artifact to not limit the ideation process. These materials serve as a medium to engage and give voice to the diverse stakeholders in taking and negotiating design decisions. Additionally, these materials help to visualize and keep track of the decisions taken collaboratively.

3.5.3 Game materials for imagination

Tools to facilitate imagining the Netherlands in 2050 By Smith et al. (2021).

This project was developed by the Transforming Practices group at the Department of Industrial Design at the Eindhoven University of Technology in collaboration with

Rijkswaterstaat, the executive agency of the Dutch Ministry of Infrastructure and Water Management. The aim was to imagine a variety of possible futures, to better anticipate and deal with the constantly changing environment in the Netherlands. Furthermore, it looked to promote discussions with a cross-section of the Dutch population, to understand how they imagine the future, what it could look like, and what it could mean for them.

The main argument behind the design concept was that imagination is a skilled activity, scaffolded by a social and material environment. This was based on an ecological and enactive perspective.

- Materials:

1. Short Speculative Stories: These are based on demographic, economic, political, socio-cultural, and technological trends. They aimed to share the trends with participants without the scientific-technological words and in a more approachable way so they could relate to them. This was through creating stories of the everyday lives of people living in 2050 and interacting with different objects for that time. The stories were recorded as 'audiobooks' and accompanied by a visual to portray the mood in which they were written.

2. Discussion tools: These tools aimed to encourage participatory sensemaking between participants. They were: A playing field and a collection of diverse objects, some recognizable, such as wooden figurines, and others more abstract and without a predefined meaning.



Figure 7. Different game materials.

How it works:

The participants listened to one of the stories, and the interviewers placed pictures in the middle of the playing field, representing the main characters of the story. Following, each participant selected a wooden figure to represent themselves and then placed it on the table expressing their relations to the characters. Then interviewers guided the participant through a collaborative imaginative discussion. During these discussions, the interviewers identified and highlighted relevant topics by placing physical objects from the discussion tools and Post-it notes on the table. In this way, the different values of the fictional actors and participants were explored to anticipate diverse futures. The discussion wrapped up with the participants identifying the three elements they found most relevant.



Figure 8. Netherlands in 2050 - Workshop

This example was selected because it not only highlights the importance of physical materials and a social setting for engaging diverse stakeholders but also emphasizes their importance as promoters of imaginary processes. In agreement with the previous example, the authors of this example, also consider that certain levels of embodiment are relevant when envisioning different aspects of future scenarios. It is also worth mentioning that it has been implemented in a real context where participants were selected from diverse population clusters. Differently from the previous example, the purpose was not to create a product or service but rather promote shared imagination of possible futures. Additionally, the setting is also different as it holds participatory discussions/interviews with fewer participants (from 2 to 3) instead of bigger workshops.

4. METHOD

This section presents the project plan and the selection of methods for each step of the design process. It starts by describing the overall methodological approach of the research. Followed by how the project was planned to fit the research and stakeholders' requirements and time restrictions. Finalized with a presentation of the design process which will be fully described in the next section.

Participatory Design is the overall methodological approach of this research. Consequently, the designer of this thesis project worked jointly with different stakeholders during the design process. A main team was consolidated to work on weekly creative sessions.

4.1 Project plan

The Re-Imaginatorium is a format that is in its development phase. At this stage, the concept and format are being tested in a set of different pilots which look to try out new advances and get feedback to feed the concept iteratively. Aling to this approach, this research is planned and structured so it fits both the academic research process and the requirements of the stakeholders involved in this first stage of the Re-Imaginatorium.

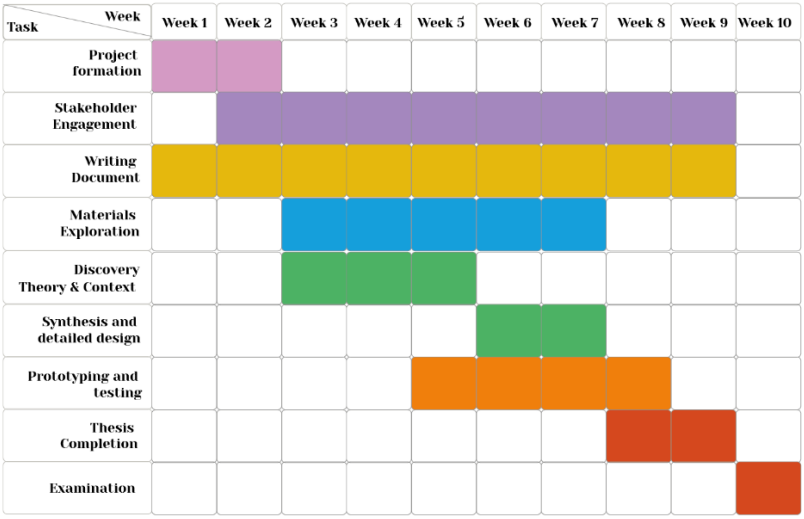


Figure 9. Research plan.

Weekly desk and on-field research were conducted. On-field research will be through different types of encounters with stakeholders. Some will be set as participatory workshops, and some others as creative and/or sync sessions.

4.2 Design Process

The design process takes as reference the general structure presented by Bobbe et al. (2016) in their study "A Comparison of Design Process Models from Academic Theory and Professional Practice" in which Analyze, Define, Design, Finalize, and Implement are highlighted as the generalized common steps among the studied models (Bobbe et al., 2016). Furthermore, this research takes, in particular, the model called "Hypercyclic" which pushes the visibility of the iterative nature of design processes, not only at macro-cycles but as well as iteration within the stages as micro-cycles (Hugentobler et al., 2004, as cited in Bobbe et al., 2016). The diagram in Figure 10 represents a sketch of the design process used in this project. This structure also aligns and responds to the ways of working and the scheduled pilots of the Re-imaginatorium.

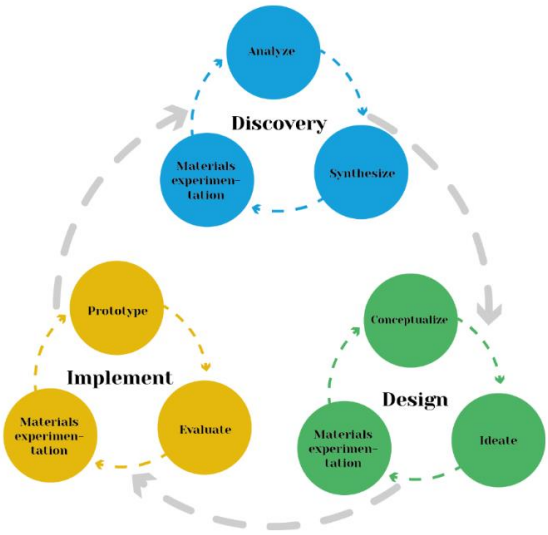


Figure 10. Design Process model

4.3 Discovery

4.3.1 Literature review

As Hanington & Martin (2019) explain, a literature review is a method of secondary research to collect and synthesize what has been researched on a given field or topic. They are not only an important part of academic research but also a useful component of any design project in both research and practice (Hanington & Martin, 2019). A literature review has the purpose of "distilling information from published sources, capturing the essence of previous research or projects as they might inform the current project" (Hanington & Martin. 2019, p.148). They also argue that the review should synthetically converge the information and show connections between references while keeping a relevant focus on the design project. This can be done by organizing the findings into categories that can be created from a chronological, thematic, or methodological perspective (Hanington & Martin, 2019). Furthermore, Hanington & Martin (2019) highlight that the sources chosen should be not only relevant to the project but also from credible sources.

4.3.2 Concept mapping

Hanington & Martin (2019) present Concept Mapping as a synthesis and analysis technique where a visual framework is created to support designers at the moment of introducing new concepts into an existing understanding of a domain so that new meaning can be made. They also state that Concept Mapping helps designers to make visible the complexity of an issue, by not only showing the existing connection but expanding what is already understood but possibly taken for granted.

Creating such a map is an iterative process in which a final map emerges when answering adequately the focus question which is the starting point of its creation (Hanington & Martin, 2019).

As statements are created through the relations represented among the different concepts, some of them may reflect knowledge that is already understood, but others will represent new knowledge. That is what Hanington & Martin (2019) highlight as the power of the Concept Map because it brings new connections to the surface within a domain that is already understood. Additionally, it supports designers to study relationships between old and new concepts and reveal new meaning in those relations.

Furthermore, Hanington & Martin (2019) emphasize the importance for the designer to have a good understanding of the domain so meaningful interconnections are generated.

4.3.3 Interviews

Hanington & Martin (2019) define interviews as an essential method to gather firsthand and personal information regarding experiences, opinions, perceptions, and, attitudes. It is not only relevant to understand users but also to collect information from experts in a particular field (van Boeijen et al., 2014). Hanington & Martin (2019) recommend to be held in person so the different nuances of personal expression and body language can be revealed during the conversation.

Interviews can be structured and or semi structured. Structured interviews follow a set of questions and are easier to manage regarding the questions and timekeeping, on the other hand, semi structured interviews allow flexibility in the conversation and are usually more comfortable for the interviewee, in any case, the researcher should be personally sensitive and have always a clear objective (Hanington & Martin, 2019).

As Van Boeijen et al. (2014) state, interviews can be relevant in diverse phases of a design process and therefore hold different purposes. In the discovery phase, they can support getting contextual information and gaining insight from people with specialized knowledge, and during concept testing, they help collect user feedback.

4.4 Design

4.4.1 Mind mapping

Mind mapping is a useful technique for analysis and synthesis when complex problems or situations are being studied (Hanington & Martin, 2019). It provides a method for visually organizing, representing, and prioritizing how different aspects of a research related to each other with the purpose of better understanding it and with the expectation of creating new knowledge (Hanington & Martin. 2019; Van Boeijen et al., 2014)

Hanington & Martin (2019) argue that as people do not think linearly and complicated issues do not follow a clear set of steps, this is a suitable method that provides "a nonlinear means of externalizing the information in our heads so that we can consolidate, interpret, communicate, store, and retrieve information" (p. 156). They also state that the mind-mapping process supports summarizing and testing assumptions, creating and dissolving connections, and thinking about diverse alternatives. In this way, "it can be used to reveal basic, idiosyncratic patterns of thinking" (Hanington & Martin. 2019, p.156).

A Mind Map can be used in different stages of the design process, not only as a tool for making sense of the information gathered but also for outlining presentations and reports (van Boeijen et al., 2014).

4.4.2 Brainstorming

Brainstorming is a method that is based on "the principle of the avoidance of premature criticism" (Van Boeijen et al., 2014, p.156) which allows to open up many types of ideas and unexpected associations, this is why it is commonly used during the ideation phase. It is usually carried out with a group of people. For brainstorming to be effective it is important to create a safe environment where participants feel free to share any thoughts or ideas (van Boeijen et al., 2014).

4.4.3 Sketching

Buxton (2007) states that sketching is core to a design process, he sees it as "a means of working through a design— sketching as an aid to thought" (p,105). Sketches hold a series of characteristics that make them open to exploration, dialogue, and feedback. Buxton (2007) listed these characteristics as follows: Sketches are quick, timely as they can be used when needed, cheap, disposable as the value is in the concept and not in the execution, and plentiful so they can make sense within a context of series. They have a clear vocabulary so it shows a sketch, a distinct gesture of openness and freedom, a minimal level of detail, and an appropriate degree of refinement. Sketches should suggest and explore instead of state confirmations and should be ambiguous, this last characteristic is highlighted because much of the sketches value "derives from their being able to be interpreted in different ways, and new relationships seen within them, even by the person who drew them" (Buxton, 2007, p,111)

Furthermore, Buxton (2007) illustrates that sketching is an activity that is about exercising the imagination and understanding, both mental and experiential, rather than focusing on the

materials used, meaning that sketches could be done with a diversity of tools and materials. This led to the understanding that sketches are also useful when designing interactions, even when they should be distinctive as they need to express unique attributes of interactive systems, they still should be consistent with the characteristics mentioned before (Buxton, 2007).

Even when sketches and prototypes are instances of the design concept, they have different purposes (Buxton, 2007). Sketches are usually used in ideation phases and as stated before they support an early exploration attitude, on the other hand, prototypes are mostly used in phases where there is more clarity on the concept and take many more resources to be built (Buxton, 2007). Prototypes will be explained in the following subsection.

4.4.4 Workshops

Conducting workshops is a central method in participatory processes (Hanington & Martin, 2019). It consists of organized and facilitated sessions for different participants to work with design team members. Hanington & Martin (2019) state that they are an efficient, compelling, and fun way to get input and build trust with the different stakeholders.

Workshops are most used in ideation phases to get insights from diverse stakeholders and generate potential solutions or ideas together. However, they also have value in evaluative sessions. In this stage, they contribute to bringing feedback and insights to review design concepts and inform design decisions for further iteration and refinement (Hanington & Martin, 2019).

A different set of materials is provided so participants can express their ideas or thoughts. The creation of a workshop demands clear structure and organization, this can be time-consuming however, the insights and results from them are highly valuable in the design process (Hanington & Martin, 2019).

4.5 Implement

4.5.1 Prototyping

Prototypes are tangible artifacts that support the development and testing of design concepts, enabling timely feedback before a product is released (Hanington & Martin, 2019). They can be materialized in different ways and using different mediums (Houde & Hill, 1997). For Hanington & Martin (2019) prototyping is a critical characteristic of a design process as it materializes what has been discovered in the research and ideation phase. They also state that there are different levels of resolution, low and high fidelity. Low-fidelity prototypes are suitable for early testing ideas and therefore are less complex, on the other hand, high-fidelity prototypes are more defined, and often communicate the final look and feel and the basic functionality. They are useful at later stages of testing to get users' input regarding aesthetics, form, interaction, and usability (Hanington & Martin, 2019).

When it comes to prototyping for interactive systems, Houde & Hill (1997) state that due to their complexity, “it may be difficult or impossible to create prototypes of a whole design in the formative stages of a project” (p. 1). That is why they have developed a model that supports the creation of prototypes that focus on exploring or demonstrating some aspect of the future artifact. It has three main dimensions: Role, look and feel, and implementation (Figure 11). They correspond to a class of questions that are relevant in any interactive system: What role will the artifact play in a user’s life? How should it look and feel? How should it be implemented? (Houde & Hill, 1997). Houde & Hill (1997) argue that focusing on what aspects

the artifact will prototype, supports more comprehensively the design process and the evaluation of particular features.

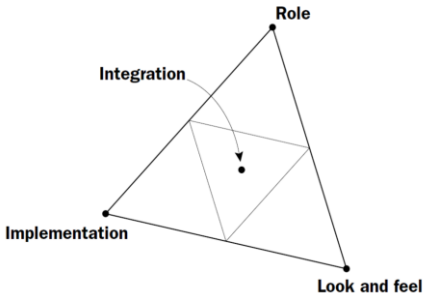


Figure 11. Houde & Hill´s prototype model.

4.5.2 Evaluative research - User testing

Evaluative research has the purpose of measuring human expectations against the designed artifact. It involves the testing of prototypes, products, or interfaces by real potential users (Hanington & Martin, 2019). Hanington & Martin (2019) argue that it is the most established form of research in design and is known as "user testing". The term has been changed to "product testing" so participants do not misunderstand that they are the ones being tested (Hanington & Martin, 2019).

This method has been expanded from focusing on performance measurements like accuracy or speed in the completion of a task, to now gathering feedback about preference measurements such as aesthetic and emotional responses from users. Therefore, it involves techniques for considering human factors and ergonomics, usability, aesthetic response, and emotional resonance (Hanington & Martin, 2019).

Hanington & Martin (2019) emphasize that evaluation should not be kept to the end of the process. Additionally, they argue that this should be iterative and based on feedback from potential users in cyclical and continuous rounds of concept and prototype development.

Furthermore, Hanington & Martin (2019) state that depending on the fidelity of the prototype, they can be introduced in real-use contexts which may bring valuable and realistic insights.

5. DESIGN PROCESS

As described in the previous section, the design process encompasses three main phases: Discovery, Design, and Implementation. The generalized cyclic flow reflects the iterative nature not only at a higher level but also within each phase. Along with the main activity in each phase, experimentation with different materials was performed.

During the whole process, participatory sessions with the stakeholders were held. Some took the form of brainstorming and ideation sessions and others were fully structured workshops. These workshops had two dimensions, one looked to support the collaborative creation of the format for the Re-Imaginatorium, and the second one aimed to test different materials. The participatory sessions were weekly scheduled and three main workshops were held during the time frame of this thesis project.

It is important to mention that as visualized in the Design Process figure, the design process was not linear and had different loops in which discovery, design, and implementation were undertaken several times.

5.1 Discovery

In this phase, the main purpose was to get a holistic understanding of the Re-Imaginatorium and stakeholders, including ways of working, needs, and expectations. Additionally, it also aimed to get a strong conceptual frame supported by previous research, cases, and relevant concepts so they provide sustain to the decision-making in the design process.

5.1.1 Literature review and Concept mapping

Secondary research was conducted through a literature review within the main field of study, Playful Design. As the nature of the Re-Imaginatorium holds strong connections with Speculative and Critical Design, these were also studied in relation to the concept of Playfulness. Furthermore, due to the goal of the Re-Imaginatorium of fostering imagination, this thesis project looked as well into the conceptualization of the nature of imagination.

The selection of literature started with the search for the basic concepts within Playful Design. First, because it was a new field of study for the researcher of this thesis project, and secondly to identify core aspects that could be integrated into the format of the Re-Imaginatorium. The main sources were academic platforms conceded to the Malmö University Library and recommendations from professors and practitioners in the field.

The main results were synthesized and structured in a concept map rather than organized by categories as suggested by Hanington & Martin (2019). It was an adjustment to the method so it could fit the design process for this thesis project. The mental map had two main intentions. On one side to visualize how concepts in each of the three fields make sense and connect among them, and on the other side, to create a framework with concrete ways of working and techniques from each one. Both were to be used in the following design phases, supporting the conceptual and practical decisions.

The relations in the concept map (Figure 12) revealed how some aspects and core concepts could work together to enhance the purpose of the Re-Imaginatorium. For example, one key common concept between Playfulness and the Critical Design Practices was ambiguity and the way it enables open or free interpretation and consequently appropriation. This in turn contributes to the higher cognitive process of imagining. Another relevant concept was the use of materiality at the moment of promoting and training imagination. This tangibility can hold thought-provoking features and also can bring common ground when in collaboration settings. Furthermore, another interesting aspect in common was the relevance given to social engagement as one characteristic that coupled with the mentioned materiality, influences the type of imagination as it helps to refine and renegotiate the process of imagination. As Hanington & Martin (2019) described as the power of this method, it ultimately supported this research by bringing to the surface new connections within the domains and revealing new meanings in those relations.

From this literature review the theoretical framework presented in section two was structured and synthesized in a way that could guide the reader in the conceptual and theoretical stands behind the proposal of this thesis project.

This literature review not only helped to make sense of the complexity of concepts but also supported the discovery of the research contribution by pointing out where this thesis project could bring new knowledge and practical evidence to established theories.

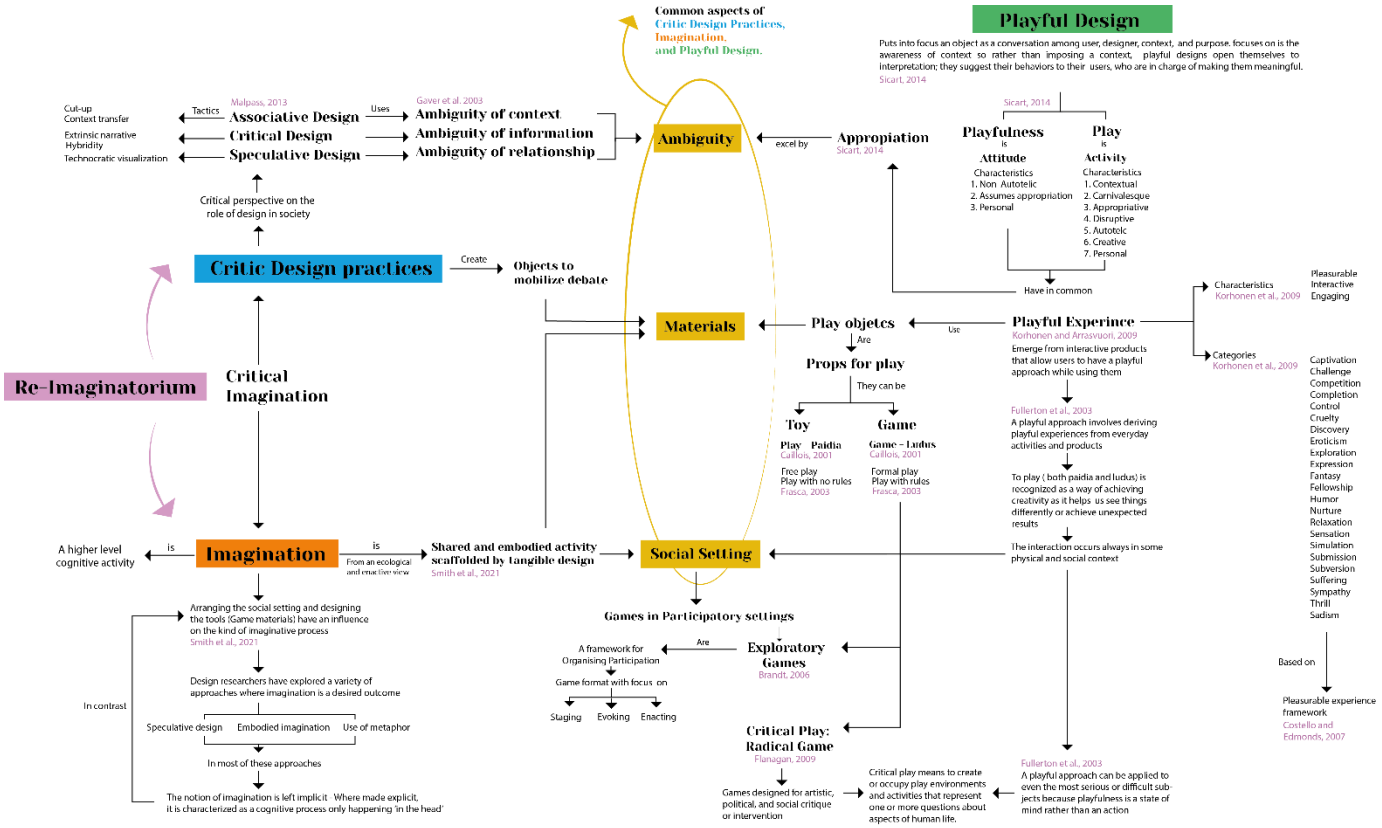


Figure 12. Concept Map from Literature Review.

5.1.2 Interviews

To get first-hand insights into the context of the Re-Imaginatorium, two semi-structured interviews were held with stakeholders. One was with Joel Veborg Co-Creator specialist within the Re-Imaginatorium. The second one was with Per-Anders Hillgren Co-creator of the Re-Imaginatorium. A third semi-structured interview was held with a professional and practitioner in the field of playful design, Daniel Velasquez Senior UX designer at Avalanche Studios Group.

Additionally, some of the fundaments of semi-structured interviews, such as questioning conversationally (Hanington & Martin, 2019), were constantly performed throughout the whole process during the weekly sessions with the stakeholders.

Two of three interviews were held in person and one in a video call. They were recorded. The majority of the weekly sessions were face-to-face. Notes were the main way of documentation.

Besides the contextual information about the Re-imaginatorium described in section two, the following were some outstanding results from the two first interviews. An important aspect was the general expectation of this collaboration. Two main objectives were agreed upon, on one hand, the support for the creation of the format and on the other hand the creation of its tangible materials. Another relevant aspect was the general approach to the creation of the Re-Imaginatorium, it holds an open and exploratory nature, not only because it is in its development phase but also because it aligns with the ontological stands of the Re-Imaginatorium itself and its creators. Therefore, the design process was also exploratory and iterative in which different ideas and materials were constantly tried out and served as a reference to build upon.

From the third interview with a practitioner of playful design within the game industry, insights about playfulness and how it can be promoted by different design characteristics were highlighted. For instance, taking LEGO as an example of the use of basic and abstract forms coupled with modularity has a significant impact on the affordances of the playfulness of a design. Another insight was that not always a playful aspect has a relevant purpose in the dynamics of a game but in the experience of the user. Different examples were exposed by Daniel in both the digital and tangible realms. To mention one, in the board game Dimant there is a cardboard "treasure box", it is a container where players can add the "gems" they win. This tangible element plays no role in the dynamics of the game; however, the users enjoy putting their "gems" there and making sound by shaking the box. This action is not only genuine but also adds a hint of competitiveness among players, as the more "gems" the louder the sound is.

These insights supported some design considerations regarding the form that the materials for the Re-Imaginatorum could have. Furthermore, brought concrete examples of how subtle details that are not directly related to the purpose of the activity can enhance a playful experience.

5.2 Design

The design phase was one in which the main group of stakeholders was actively involved. In this phase, the main purpose was to make decisions about the format of the Re-Imaginatorium and which and how the materials should be configured so they support the activities and ways of working within that format.

5.2.1 Brainstorming

Brainstorming was one of the main methods through which the weekly sessions were undertaken. They often had an exploratory approach and were open enough to let the team share any type of idea or thought. Their purposes varied, however, they can be categorized as related to three main purposes: Developing the format, the content for the pilots, and the materials.

These sessions were from one to two hours long, included three to four participants, and used basic materials like pens, paper, and a digital board in Mural (Collaborative digital platform) for documentation.

It is worth mentioning that while brainstorming about the theme, which for the first pilot was "Re-Imagine Universities" the team tried to imagine together how universities could be. This in a way was enacting a basic "Re-Imaginatorium" process that brought important insights about which challenges others could face, which tools could be needed to support the re-imagination process, and which aspects helped us to re-imagine the universities. This in turn revealed relevant aspects to take into consideration in the format and the material.

The main results related to the format were: The need to have different moments when re-imaging, they should be flexible to be adapted in different contexts and with different themes. For the first pilots, three main moments were selected. One that serves as an introduction and inspiration so participants can land on the theme and get different references and examples. Another was to evoke so participants could warm up their imaginations by starting from broad and even unreal perspectives. The third one had a creative approach so participants could formulate more concrete ideas and visions about the theme. Even when these three moments seem to be divided, they overlap and build upon. Relevant aims were to promote exploration, reflection, and creation in the participants around the theme. The format is still under construction but these considerations guided the structure of the first pilots.

Accordingly, the materials should host qualities to enable these two moments. Details will be described in the prototype section. Some generally required characteristics for these materials were to be evocative, serve as a medium of expression, and enable scenario-based creation.

The results from the brainstorming in the weekly sessions were essential throughout the complete design process, as it not only informed each decision but also was the main method for collaborating with the stakeholders.

5.2.2 Mind Mapping

Mind mapping was a tool constantly used throughout the whole process. Based on the results from the brainstorming and other sessions with the stakeholders, mind mapping was a method that supported making sense of the information collected, synthesizing it, and prioritizing it. Furthermore, aligned with Hanington & Martin´s (2019) arguments, mind mapping helped in a non-linear manner to express and share with stakeholders the structure and the rationale of the design decisions regarding the format and materials.

From the synthesis of data done through the different mind maps relevant information was pointed out. An important example of it was a series of concrete tactics and categories that other authors have used when designing, on one hand with the concept of ambiguity and on the other hand playful experiences. The wheel in Figure 13 combined them, each level displaying a cluster of information. The inner circle contains a taxonomy of ambiguity (Gaver et al., 2003), the middle circle the tactics used in critical design practices related to that taxonomy (Malpass, 2013), and the outer circle the set of categories of playful experiences (Korhonen et al., 2009). Appendix 1 shows in detail the three frameworks made for each author.

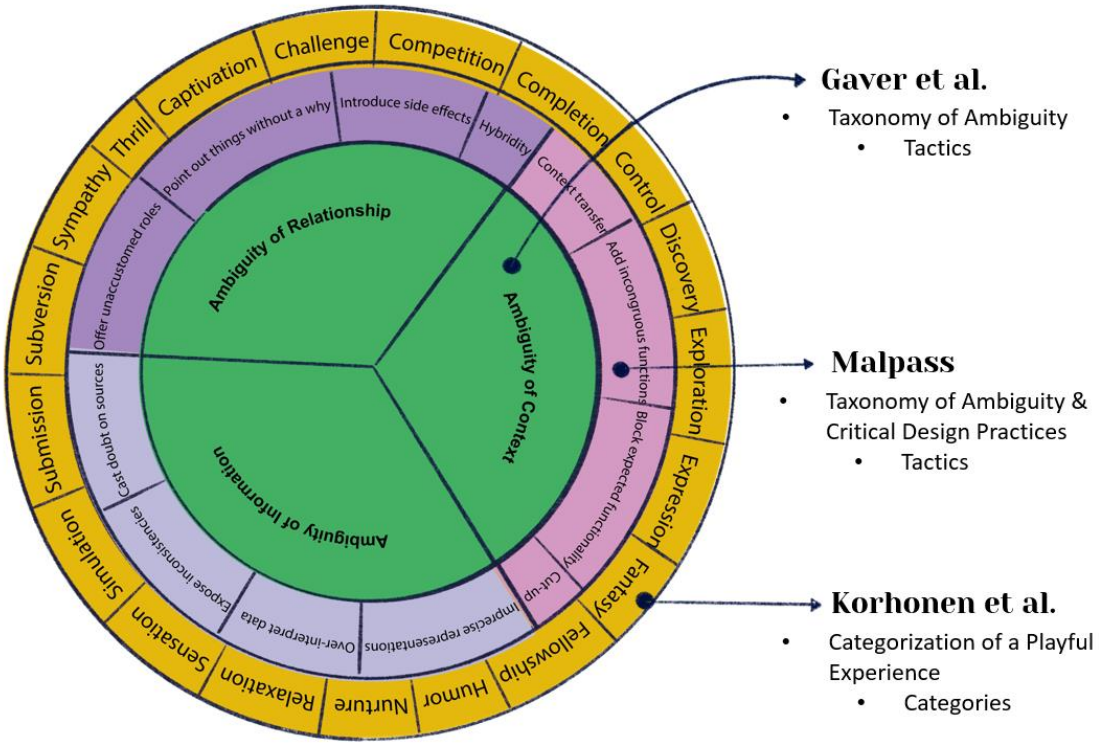


Figure 13. The tool that supported the design of each piece of the materials.

Each piece was the result of rotating and playing around with the three levels of this wheel, creating different combinations among them. These random combinations prompted brainstorming and consequently supported the conceptualization of each material at the design stage. An example of a prompt was: “How to create a material that reflects ambiguity of context (inner level), by adding incongruous functions (middle level), and creating a sense of fantasy (outer level)”. Some examples of materials created with this wheel will be described in the next section.

5.2.3 Sketching

Agreeing with Buxton (2007) sketching was a core part of the design process. It supported the thought process at the moment of generating ideas for the format and materials as well as being the starting point of conversations and discussions about how those ideas might fit better the Re-Imaginaorioum.

Sketches took the shape of hand-drawing graphics, digital visuals, and mixed depending on the intention of the communication. Some were used as a personal thought-process tool and others for sharing with stakeholders. It was also a tool to see which concepts and aspects of them were relevant to continue to work with and which others had less relevance or impact in the general purpose of the Re-imagnatorium.

5.3 Implement

This phase was also highly exploratory. The implementation in the time frame of this thesis project was embedded in a set of pilots to inform the development of the Re-Imaginatorium format and materials rather than finalize it. These pilots were structured into workshops and used different sets of prototypes of the materials.

5.3.1 Prototyping and workshopping

Three prototypes were created and tested with real users. All of them involved testing the design concepts for the format and the materials. Two were in person and one was online, with the purpose of testing how different platforms affect the experience and results of the Re-Imaginatorium. The theme in these pilots was “Re-Imagine Universities” accompanied by three sub-themes: Health, Superdiversity, and Sustainability.

Based on Houde & Hill's model (1997) the prototypes focused on exploring and demonstrating aspects related to the role and the implementation. For example, the materials created were intended to test how the concept of ambiguity reflected in the diverse tangible features fulfills the role of enhancing imagination. Likewise, the combination of the structure, tone of voice, timing, and main activities of the format were tested to understand how they work together so it could be used and adapted in the future.

The format:

The three pilots tested slightly different configurations of moments in the format, however, three main moments were common in all of them: Contextualize, evoke, and re-imagine. The first moment had the purpose of bringing context through examples. The second moment was intended to warm up the imagination process. Finally, the third moment looked to re-imagine the universities from a more concrete perspective.

The tone set in the workshops was very flexible and open. The activities had a minimum of rules, of which the majority were optional so participants could follow them or not as they fit into their imaginative and creative processes. Activities and materials for each workshop are described below.

PROTOTYPE ONE

The first workshop was two hours long. Participants were students and professors of the Humboldt University. They were divided into three groups, each one corresponding to one of the sub-themes. The workshop was conducted in person at the Humboldt University. The workshop was facilitated by Professor Per-Anders Hillgren from Malmö University.

The Workshop activities:

1. Contextualize
   This was an introduction to the workshop and a presentation of inspirational examples, in this case, some of the radically re-imaged universities by Staley in his book *Alternative Universities: Speculative Design for Innovation in Higher Education*. (Staley, 2019).

2. Evoke
   After having heard and seen examples of radical versions of universities, the participants were given the task of creating a short story or description of "a university" based on a set of cards, randomly selected with evocative visuals and texts.
   The activity instructions were:

   Purpose: To warm up our imagination by allowing us to make connections where they might not have been seen otherwise.
   It is allowed to think of the craziest or silliest ideas. Dream big!

   - You will get a set of cards that have been selected randomly. They have inspirational images and a couple of words to suggest meanings to you.
   - You can arrange them as you wish to **create a short story** about "a university".
   - There is one card (the yellow one) in which you might write an adjective that you select from the list, and think about its hyperbole (the exaggerated version). This you might use to qualify "the university you will create". This can be done at the beginning or the end, depending on how it fits best with your creative process.

   Help: The cards have different colors that respond to the categories of how, what, where, and who, but you are free to use them as a guide or not.

   Tip: You can take notes while creating "your university" so they can support you when sharing it with the other teams.

3. Re-imagine
   Once participants entered an open space where all could be possible, they were guided into a more concrete discussion. The activity consisted of Re-imagine the universities through the lenses of one of the sub-themes. The instructions were:

Purpose: To re-imagine the university
As with the previous activity, It is allowed to think of the craziest ideas. Dream big!

Now that you have warmed up, it is time to re-imagine what universities might look like in the future.

- You will start by selecting a topic from the list. It will act as lenses through which you will reflect on the different aspects of a university.
- You will receive a blank set of cards for you to write, draw, sketch, or use as you might fit better in your creative process.
- The cards have written the categories of how, what, where, who, and why for you to reflect upon. However, you are free to come up with different ones.
- Additionally, we have prepared a set of questions/thoughts for you to guide the reflection on each of the above categories.

Based on those reflections **you will create a narrative or description** so you can share it with the other teams. You can use any of the materials to express your ideas.

The materials:

One of the main purposes of the creation of these materials was to test the concept of ambiguity as a tool for promoting open interpretation and therefore appropriation, which is one relevant characteristic of playfulness. Consequently, to test how in the interaction with those materials participants can have a playful experience.

1. Contextualize
   The main material was a set of examples of radically re-imagined universities. They were presented on a screen accompanied by a verbal description with reflective and open-ended questions about them. This introduction was very similar in all three prototypes.

2. Evoke
   The main resource was a set of cards that displayed images specifically created to evoke reflections about what, how, where, and who. Figure 14 displays the rationale behind the category of the cards.

Trial of materials 1

Type of cards	Category	Examples
Basic Characteristics	• **Where** learning is happening. Images of diversity of places/spaces. • **What** type of knowledge. Images with different subjects.	• **Where:** Mobile ship, permaculture ecosystem, top of a mountain, playful room, experimental lab. • **What:** Food in the future, urban planing, music, performance, human design.
Special Actors and special characteristics	• **Who:** Different type of humans and non-humans • **How:** Ways of learning	• **Who:** Kids, elderly, animals, non-humans, humans with special abilities. • **How:** Embodied, attentiveness,multi sensorial, mixed reality.
Actionable Participants to fill in	**Giving a name to the University** 1. Participants select a adjective 2. Write it on one half of the card 3. Then find its hyperbole 4. Write it on the other half of the card	**List of adjectives:** Adventurous, calm, fragile, cloudy, dizzy, crazy, doubtful, fanny, impossible, muddy, odd, open, strange, tasty, thoughtless, playful.
Random Adds on	Random cards to add complexity to the description and possible unexpected combinations	

Figure 14. Categories for materials of the first prototype.

The visuals, words, and other elements were inspired by the results of the brainstorming sessions with stakeholders and guided by different combinations in the wheel (Figure 13). The images were created with Midjourney. Appendix 2 shows some of the sets of images created. Examples of how some of the cards were created are explained below and visualized in Figure 15.

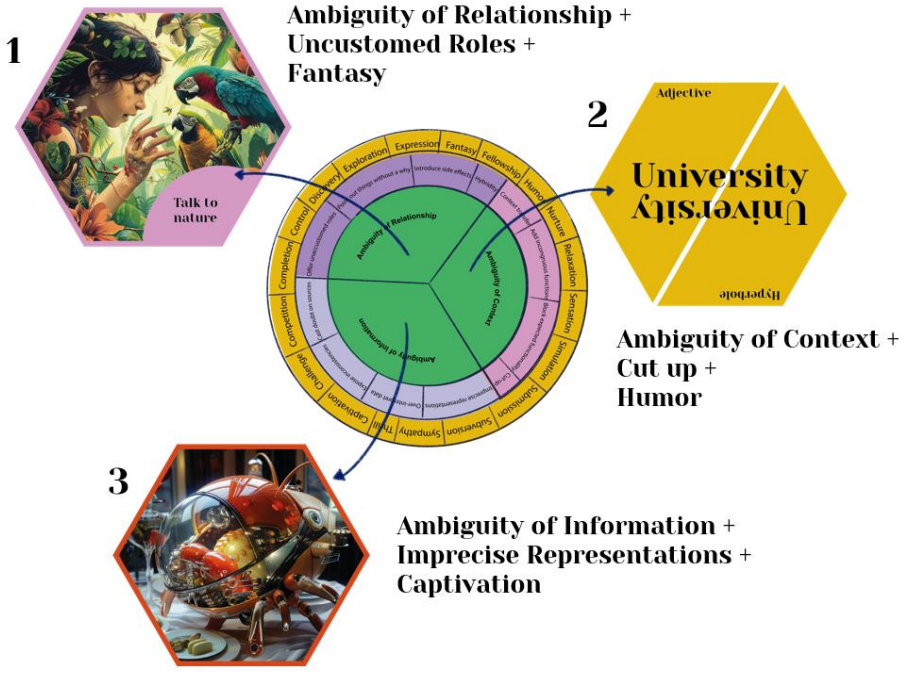


Figure 15. Three examples of combinations in the wheel and their resulting cards.

1. The first card was guided by the combination: Ambiguity of Relationship + Uncustomed Roles (tactic) + Sense of Fantasy (Playful experience category) + Sustainability/More than Human/Different world views/Attentiveness (Themes and concepts from brainstorming sessions). This type of card has visuals and words, which lead to a more focused reflection. However, interpretations could still vary significantly as the ambiguity set by the unexpected role of "humans able to talk to nature" opens up the spectrum of possibilities and capabilities of nature, including humankind. This in turn, potentially resembles a fantastic world.

2. The second card was guided by the combination: Ambiguity of Context + Cut-up (tactic) + Sense of humor (Playful experience category). In this case, the cut-up tactic was used only with words. Here adjectives and their hyperboles were introduced into unexpected contexts as they had nothing to do with the university's common vocabulary. The purpose of this card was to be used to name or qualify the university participants were asked to imagine. One of the adjectives, for example, was "Claudy", when combined the substantive "University" serves as a way to change contexts and generate reflections about what "A Cloudy University" could look like, how it would work, who would be the students, etc. Furthermore, this incoherent qualification could potentially have a humorous tone.

3. The third card, was guided by the combination: Ambiguity of Information + Imprecise representations (tactic) + Captivation (Playful experience category) + Super Diversity/Food as cultural expression/future (Themes and concepts from brainstorming sessions). This card, as part of the random card type, aims to express unclear and eye-catching information to capture the participant´s attention. This is with the purpose of promoting imagination through the dialogue about what those visuals are or represent, what that could mean, etc.

The shape of the cards was hexagonal allowing different configurations (linear, centrical, branched, etc.) so participants could arrange them in the order they fit better their thinking and creative process, as well as better represent the relations among them.

Additional resources like Post-Its, paper, and color pens were also provided.



Figure 16. Example of a set of cards.

3. Re-imagine
   Here also the main resource was a set of hexagonal cards but at this time they were black for participants to write, draw, and sketch on them so they served as a medium for organizing ideas and visualizing them to share with other participants. They had the same color code used in the previous activity to guide reflection about the different aspects of a university. In this activity, participants could use them as well as the previous cards as they fit into their creative process and ideas to communicate.

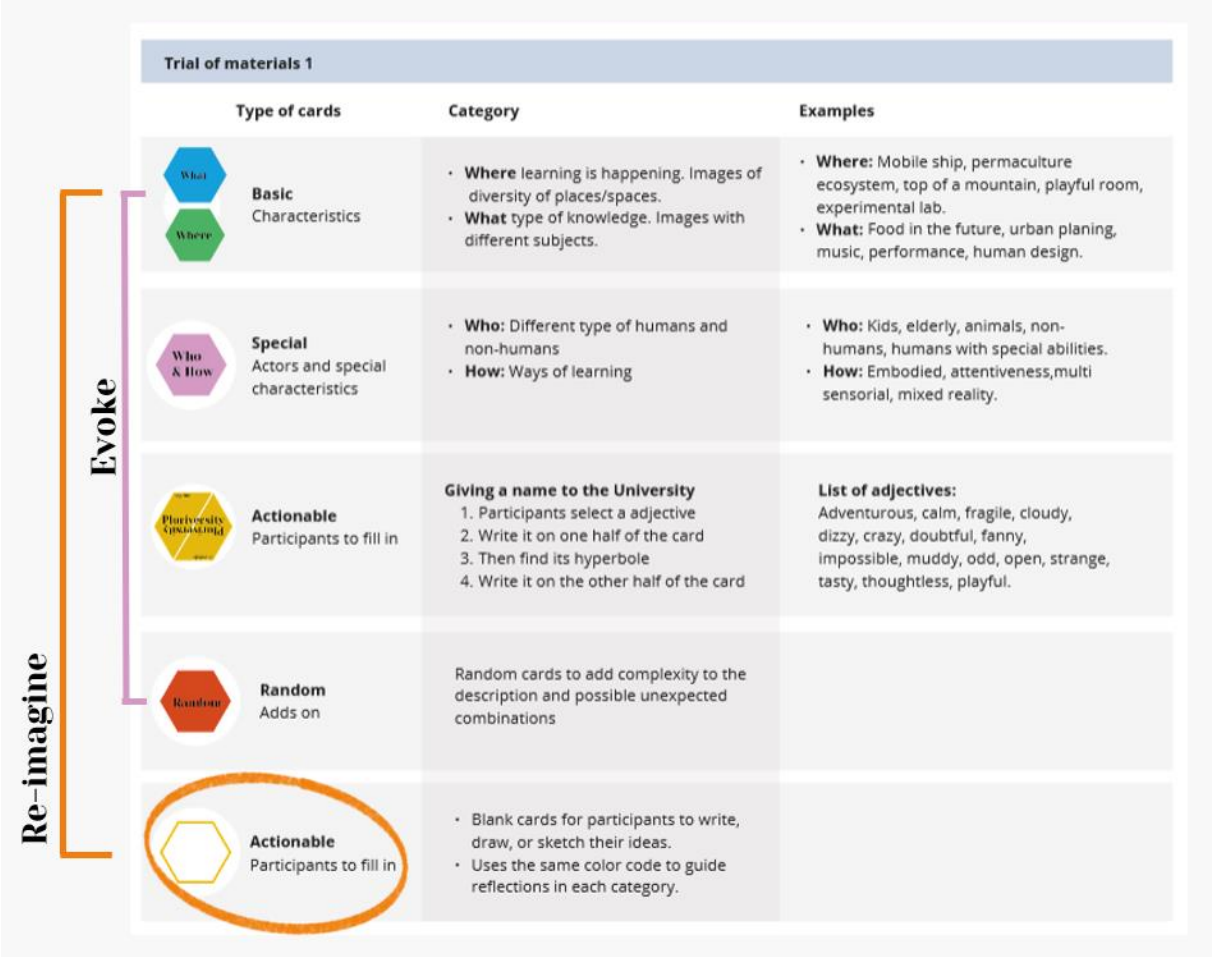


Figure 17. Type of cards used in the different activities. The orange circle shows the added blank cards for the Re-Imagine activity.

PROTOTYPE TWO

This workshop was conducted online, and the participants were situated in five different countries of Europe. Participants were representatives from different universities of the UNIC alliance. The workshop was divided into two sessions of two hours each. Seven participants were divided into three groups each one corresponding to one of the sub-themes.
It was conducted via Zoom and the collaborative platform used was Mural.
It was facilitated by professors Per-Anders Hillgren and Per Linde, co-creator specialist Joel Veborg, and Master´s Student Grace Uribe, all from Malmö University.

The workshop activities:

1. Contextualize and 2. Evoke moments were the same as the previous one. However, the 3. Re-imagine was slightly different.

3. Re-imagine
   In this version, in addition to the activity from the first workshop, participants were asked to come up first with three to four ideas of possible "universities" to after elaborate further in one.

The materials:

1. Contextualize
   The same as the previous one.

2. Evoke
   The same set of cards as the first prototype, however, they were placed on a board at the collaborative and digital platform Mural. This is because this workshop was run online. Besides this, participants were free to use the different tools from the platform, such as post-its, icons, and text boxes, among others. See Figure 18 which is a screenshot of the created environment.

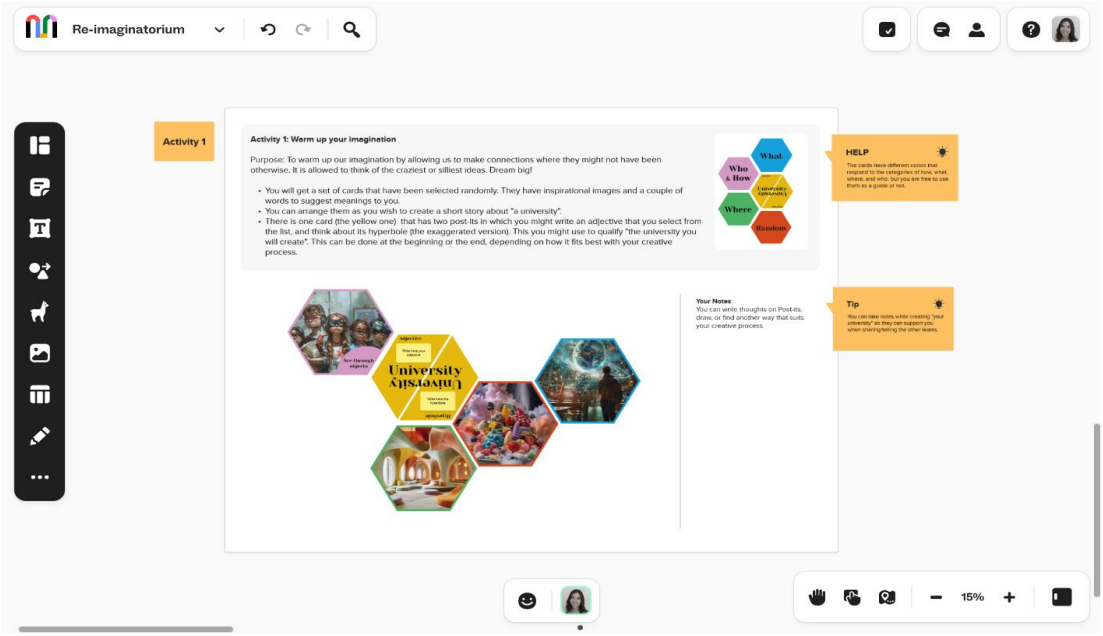


Figure 18. Screenshot from the workshop environment in Mural.

3. Re-imagine
   In this case, instead of blank cards participants got a blank board with four divisions for each of the possible ideas. Participants were encouraged to use the different materials given, the Mural tools, and other digital resources to express their ideas.

PROTOTYPE THREE

This workshop was conducted in person at the Malmö University. Participants were 18 members of the student board of UNIC, all from different universities and programs around

Europe. They were divided into three groups, as in the previous workshops, each one corresponding to one of the sub-themes.

It was facilitated by Professors Per-Anders Hillgren, Joel Veborg, and Master´s student Grace Uribe, all from Malmö University.

Workshop activities:

1. Contextualize
   The same as the previous one.

2. Evoke
   Based on the previous experiences, ambiguity in the visuals coupled with the dynamic of the main task of the Evoke Moment (Building meaning out of random and unexpected combinations of words and visuals, in a way that could be the source of inspiration for the story participants were asked to create) demonstrated to have potential to foster imagination. Accordingly, it was decided to upgrade the activity by expanding the range of human senses to be used in it, allowing a more embodied experience. Therefore, ambiguity could be implemented not only in visuals but, for example, in smells, sounds, textures, and their combinations.

   Even when the developing team called this trial workshop "a light version of embodiment" it added extra sensorial layers with the purpose of opening up imagination from a multisensorial aptitude. Furthermore, this is a feature that potentially enhances a playful experience (Korhonen et al., 2009).

   Building up from the sight sense that the previous materials primarily used, three more senses were incorporated: Hearing, smell, and touch. The participants were divided into three groups, each one got one of the added senses.

   The main purpose of the activity was the same as in the previous workshops: To create a story or a description of a university based on the materials received. This time with new pieces that held different sounds, smells, and textures.

3. Re-imagine
   The main activity was also similar to the previous workshops, to re-imagine the university by reflecting on its different aspects. However, this time participants were encouraged to materialize their ideas into scenarios or other types of compositions that represent the different actors, elements, characteristics, and the relations among them. As in the evocative activity, here it was also added the additional senses to build upon when re-imagining the universities. Furthermore, additional pieces with diverse shapes were incorporated to support the construction of these scenarios.

The materials:

1. Contextualize
   The materials were the same as the previous workshops.

2. Evoke
   New pieces with sounds, smells, and textures were incorporated for participants to reflect upon and make sense of them in relation to the other elements used in the previous workshops.

The selection of the smells, sounds, and textures was based on the feedback from previous workshops, the results from the brainstorming sessions with stakeholders, and/or the personal experiences of the stakeholders' team and the researcher of this thesis project. The general rationale behind the selection was to have familiar and non-familiar elements that could evoke different sensations, thoughts, memories, and reactions.

- Sound cards:

There were eight sounds selected: A person whispering, sounds of space, fair scene, farm sounds with light water trickle, energic crowds and happy reaction, clock ticking, party horn, and powerful thunder. The sounds were to be discovered by interacting with a board (Figure 19). The different hexagon shapes on the board had different sounds that were activated by placing one of the individual hexagon cards on top of them.

The board was created with a Bare Conductive Touch Board. This microcontroller board has dedicated capacitive touch and MP3 decoder ICs, it is based around the ATmega32U4 and runs at 16MHz from 5V. It has 12 capacitive touch electrodes, from which 8 were used for this workshop with each of the mentioned sounds. Conductive tape was the main medium for creating the circuits. They were put on the back of the paper board.



Figure 19. Sound cards being placed on the different fields of the board.

- Smell cards:

There were eight smells each on a card. The essences were cocoa, apricot, butter, celery, pepper, licorice, and animals. For each smell card, a sachet was made with a plastic mesh containing perlite impregnated with the corresponding essence. Then these sachets were placed inside each of the cards (Figure 20).

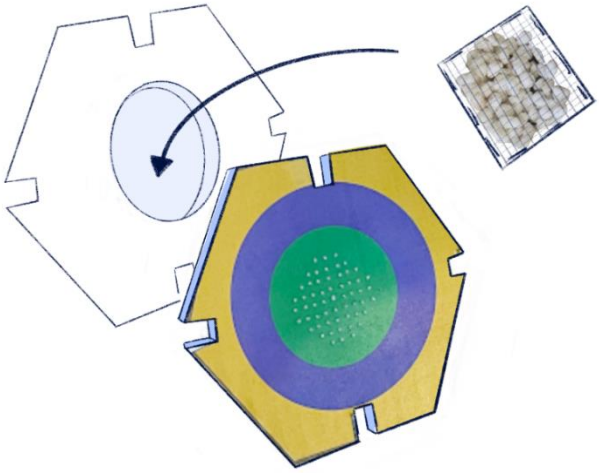


Figure 20. Construction of the smell cards with a sachet containing perlite.

- Tactile Cards:

Different materials were selected to bring the tactile experience. For example, elastic silicon was selected to evoke a feeling of flexibility, and wood to evoke nature and warmth. In total, there were eight different materials providing textures based on their physical characteristics. They were wool, aluminium, silicon, wood, adhesive plasticine, double-sided tape, cork, and polyethylene. They had the purpose to evoke sensations such as softness, stiffness, flexibility, stickiness, prickly, etc. However, it was understood that what they could evoke in each participant was going to be based on their personal experiences.

The figures below show the cards created for the workshop.

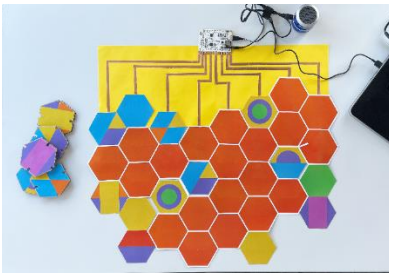


Figure 21. Sound cards and board



Figure 22. Tactile cards



Figure 23. Smell cards

3. Re-imagine

Likewise, new pieces were added to this activity. These had an expanded range of shapes so that participants could appropriate them and assign meanings as they felt they best fit to represent the elements of their ideas. Additionally, the pieces were modular and could be ensembled together in 2D and 3D. It gave an extra dimension of expression and visualization. These two characteristics had the purpose of facilitating the construction of tangible scenarios, where different actors, relations, and other elements could be represented through the pieces.

A basic example could be a scenario that wants to showcase the collaborative work between students of a university and the municipality workers, it could be expressed by the symbolic representation of two people/actors (students and workers accordingly) created with a triangular piece as a base and a circular piece fitted into the top. This could be then placed on top of a hexagonal piece that symbolizes the university and connected to another hexagonal piece that symbolizes the municipality. Indicating that the university and the municipality are connected and work together.

As in the previous workshop, the pieces used in the "Evoke" moment could also be incorporated into the composition since the shapes of those pieces were part of the modularity system. Therefore, sounds, textures, and smells could be also part of the scenario created.

The figure below shows the added cards on the Evoke and Re-imagine moments and their characteristics.

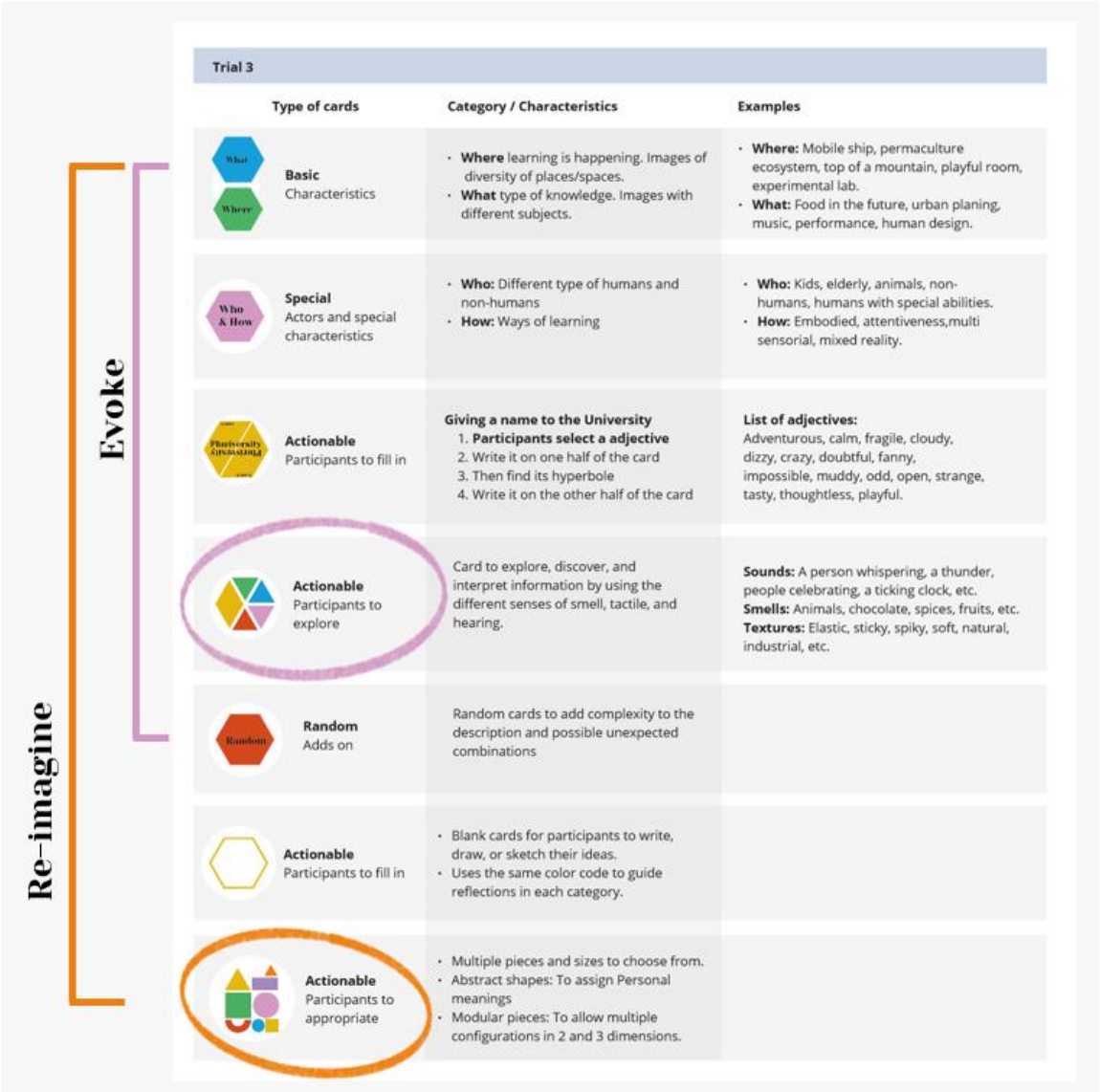


Figure 24. Type of cards used in the third workshop. The purple circle shows the added cards for the Evoke Moment. The Orange circle shows the added cards for the Re-imagine Moment.

Examples of how some of the added cards were created are explained below and visualized in Figure 25.

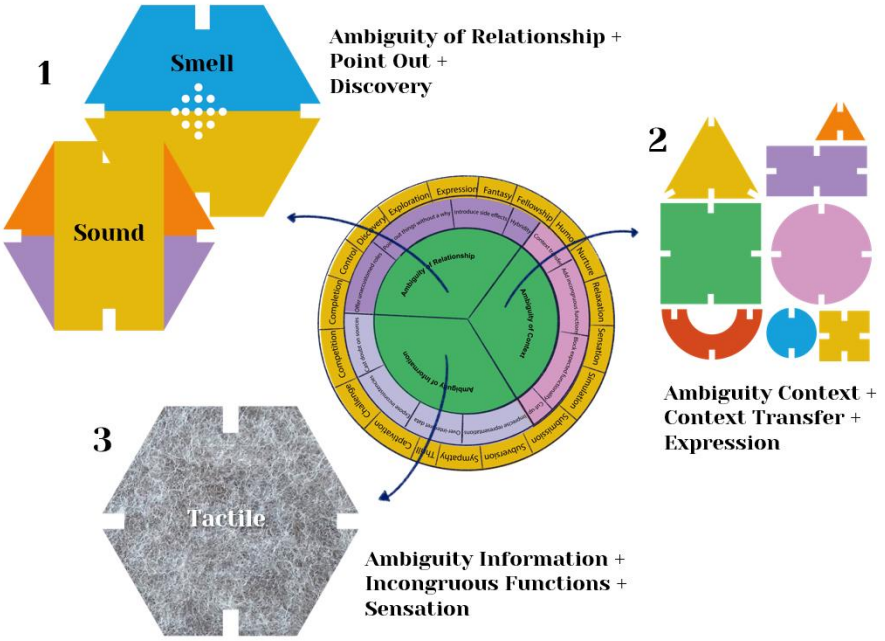


Figure 25. Three examples of combinations in the wheel and their resulting cards

1. The first example was inspired by the combination: Ambiguity of Relationship + Point Out (Tactic) + Discovery (Category of a playful experience). As usually smells and sounds evoke personal experiences and memories, they were a perfect medium to promote Ambiguity of Relationship, which is about personal stand about a situation. By pointing out one smell or sound in each card, participants can get time to focus and reflect specifically on it. Additionally, as the essences and sounds have no label, they are open to multiple interpretations and promote a sense of discovery while trying to guess or make sense of them.

2. The second example was inspired by the combination: Ambiguity of Context + Context transfer (Tactic) + Expression (Category of a playful experience). In this particular case, this combination supported the thinking behind the pieces for facilitating the scenario building at the Re-imagine moment. The abstract, modular, and basic shapes have the purpose of providing a medium to make tangible ideas by assigning meanings to each piece. For instance, a piece could symbolize an actor or a place, and its position relative to other pieces could indicate the nature of their relationship. These symbolic representations would transfer the context of the participants' ideas into these basic shapes, that would only make sense in that configuration. Additionally, diverse shapes and sizes are provided to participants to use freely as a means of personal expression.

3. The third example was inspired by the combination: Ambiguity of Information + Incongruous functions (Tactic) + Sensation (Category of a playful experience). Here, the tactile aspect is highlighted as the different textures can visually stand out from the other cards, introducing ambiguity about their function or the information they convey. Furthermore, it adds another sense to the sensory experience.

It is worth mentioning that some of the tactics and the playful experience categories overlap, meaning that they are not exclusive to only one combination. For example, the cards with smells not only support a sense of discovery but also sensation and exploration.



Figure 26. Three sets of materials. Containing the cards with the visuals and some of the geometric modular pieces.

Furthermore, all materials were crafted in cardboard and paper. This had the initial intention of facilitating its replicability, transportation, and use by other people in other countries. The reason why paper circuits were the main technology used. Below are some pictures of the material experimentation with paper circuits.

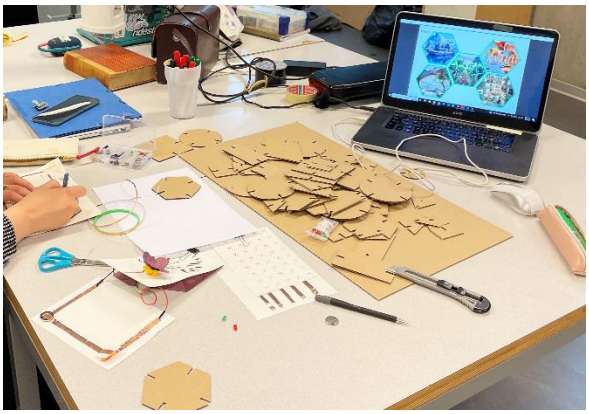


Figure 27. Experiment with conductive tape and different components, such as light, sound, and motion.

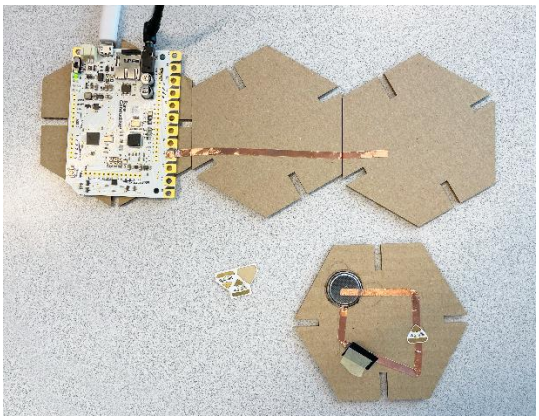


Figure 28. Experiment with a touch board and conductive tape.

5.3.2 User testing

User testing was coupled with prototyping, both performed in the workshops with real users. Furthermore, as Hanington & Martin (2019) suggest this was not done solely at the end but rather along the process in cyclical rounds of concept and prototype development. Aligned with Hanington & Martin's (2019) suggestions, the workshops not only focused on measuring or qualifying the accuracy in the completion of the tasks on each activity but also the emotional responses of participants. With this approach, the following were the main results collected through the feedback from participants after each workshop and observations during the

workshops from the stakeholders involved and the researcher of this thesis project. Additionally, results were evaluated from the quality of the re-imagined universities proposed by participants at the end of the sessions.

First workshop:

- Engagement: The overall participation among participants was high. This was demonstrated not only in the quality of the results but also in the fluid dynamic during the sessions. However, it was difficult to zoom in on each group to observe the ways of engagement as there was only one facilitator. Therefore, there is a lack of empirical evidence.

- Affordance of materials: The sets of cards were used as intended. The combination of words and visuals brought together unexpected reflections and ideas. The words "what, how, where, why, and how" were perceived as useful categories at the moment of reflecting and re-imagining.

- Playful experience: General feedback from participants indicated that they had an enjoyable experience. Curiosity and intrigue were some of the main attitudes in the sessions. For example, one participant commented "There was an intriguing and thought-provoking discussion about the adjective "doubtful" and what it could mean in a university".

  A sense of freedom and openness was also generally experienced. A participant commented, "I really liked it, an open space to talk". One feedback from a stakeholder stated, "It was allowed to take a crazy idea and play with it, and then try to see how it possibly could make sense or how it might be relevant ... going back and forward with the materials, that was very playful".

- Quality of the results: Based on stakeholders' reflections the results were very strong. They formulated very unexpected ways to see the universities and their different aspect.

  An example is the Night Radio University. This is a university that runs its classes at night and where learning is centered around sound as a medium. Some aspects are: Combining music and academic knowledge. The special atmosphere at night brings attention to existential questions that might be difficult to talk about in the daytime. Listening sharpens when it is dark and students learn how to engage in attentive and active listening. Dreams play an important role in discussions. Among others.

Second workshop:

- Engagement: The overall engagement in the two digital sessions was high. All participants actively participated in the different activities and contributed to the main task. Participants have commented in the feedback that this engagement could have been due to diverse aspects and/or combinations of them. One of the highlighted aspects was the adequate peace of each session, allowing enough time to reflect and create. A participant commented that usually in collaborative and creative workshops there is a sense of rush in generating ideas fast, this contributes to making some people feel stressed and therefore refrain from participating.

  Another highlighted aspect was the ability and competence of the facilitators regarding the theme "Re-imagine universities". Participants noted that knowing about the subject supported the flow of the reflection, for example by questioning when needed to spark reflections.

An additional aspect that contributed to the overall engagement, was also related to the small size of the groups. It allowed participants to easily engage with the materials and communicate with the other group members and with the facilitator.

- Affordance of materials:

  From feedback and observations, the materials were easy to use, it can be argued that it was due to the fact that they were supposed to be used as they fit better the creative process of each group, therefore there were no right or wrong ways of using them.

  The radical examples of re-imagined universities plus the evocative images and words were highly appreciated by the participants. For example, one participant commented, "Setting the scene by presenting very radical examples was crucial as it allowed us to think also very radically and big". Another comment was "There was something there when thinking about universities in relation to the adjective fragile that really made me think differently about them".

- Playful experience:

  The participants stated that the workshop was enjoyable and they got the feeling of gaining something out of it, on a personal and also on a team level. The diverse way in which participants used the materials to create their visualizations, revealed that appropriation was performed.

  Three main categories of playful experiences were identified, exploration, expression, and calmness. Exploration was mainly perceived at the evocative moment when participants interacted with the cards with images. The ambiguity of the visuals in some of the cards allowed multiple interpretations that were a central point for starting a conversation among participants in each group. Expression was the ultimate experience, most participants felt that they contributed to the creation of something. Finally, The experience of relaxation or callmess, was one unexpected result. It was related on one hand to the peace of the activities in which participants did not feel in a rush. On the other hand, due to the general sense of freedom created in the atmosphere, it was allowed and even encouraged to imagine "crazy" or "silly" ideas.

- Quality of the results:

  The three groups had achieved the main goal of re-imaging universities, some about general characteristics of a university and others in specific zoom-in aspects of them. Likewise, some ideas were more abstract, and others more concrete. In general terms, the main team of Stakeholders considered that the participants had profound reflections about diverse aspects of a university and came up with interesting ideas.

  Below is an example of a concept/idea from a team that reflected on universities through the health sub-theme. They chose the mental health of students, professors, and other actors in the university and created, what they called, The self-awareness lobby. This idea is about having a lobby that when people enter the university helps to self-reflect on how their bodies and minds are on that day and find a route that leads them to different spaces at the campus that can support that state (Ex. need for focus, need of socialization, among others).

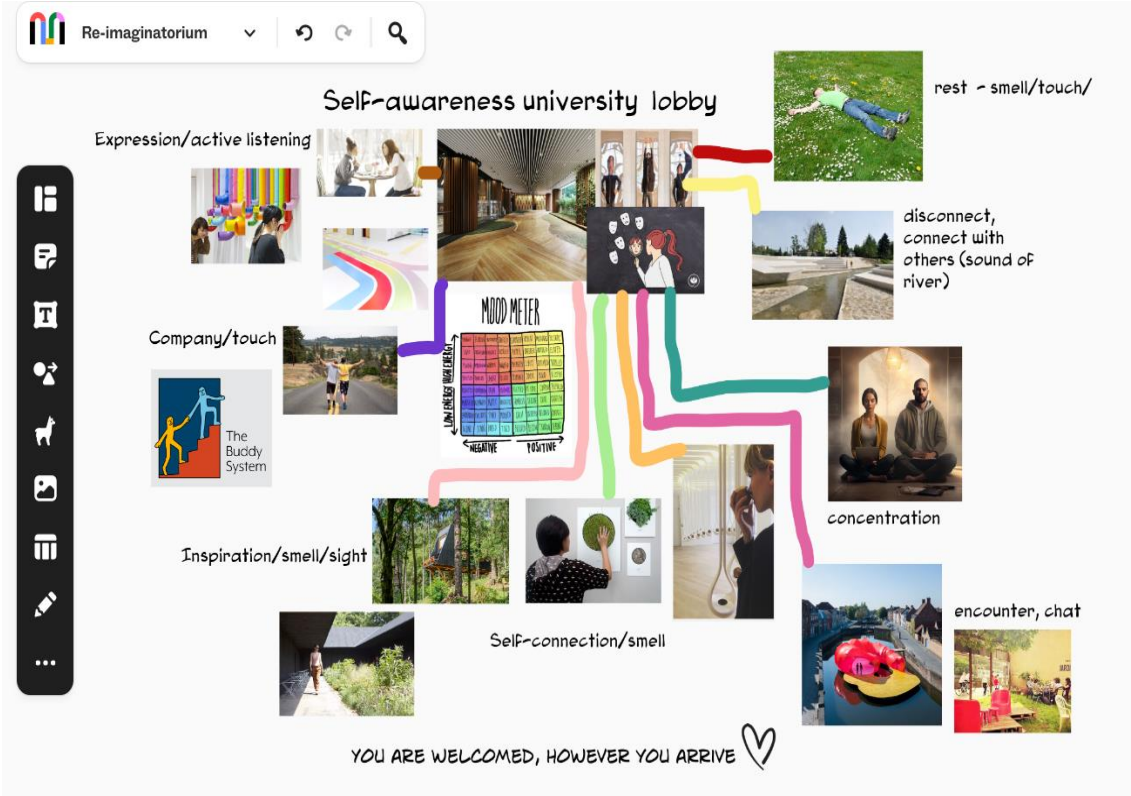


Figure 29. Example of one result from the Re-Imagine activity.

Third workshop:

- Engagement: In this particular workshop it was clearly recognized and observed the high engagement of participants. All participants actively participated in the different activities and enthusiastically interacted with the different materials, other participants, and facilitators.

  Even when this workshop was the one with more participants, arranging the teams and enabling participation was relatively easy for the facilitators.

  Several participants expressed their motivation to participate in further Re-imaginatoriums, and even some reached out to the facilitators to discuss possible ways of collaborating.

- The affordance of materials:

  In this workshop, the physicality of materials played a central role in the way participants engaged in the reflections. Even when the materials had different characteristics for each activity, their modularity made it easy to understand the continuity of the activities and supported a smooth transition between them.

  For the evocative moment, the use of the different senses proved to enhance not only the experience but also the ability to find connections where they would have not seen otherwise. In the Re-imagine moment, materials allowed participants to build together a physical representation of their ideas. However, even when the ideas were quite concrete, the participants created abstract representations that went further from the scenario-based intention of this activity.

- Playful experience:

  Most participants stated that they highly enjoyed the workshop. The materials were appropriated in very different ways in the three groups, it was demonstrated in the final representations of their Re-Imagined universities ideas.

  Three main categories of playful experiences were identified, exploration, sensation, and expression. As in the previous workshop, exploration was mainly perceived at the evocative moment when participants interacted with the different cards. Furthermore, in this iteration, the extra cards with the different senses add a sensory layer to the experience. The ambiguity of the visuals, the unexpected combination of words, and the different sensorial inputs teamed up in a way that allowed multiple interpretations and sparked deep reflections among participants. Even when some participants shared their skepticism about how the use of different senses could work in that context, they shared after the session that they found it quite relevant and that they positively contributed to the activity's purpose.

  Expression was another highlighted experience, some participants commented that they felt they could express themselves, not only verbally in the open and safe atmosphere of the workshop but also through the materials when materializing their ideas.

- Quality of the results:

  The main team of Stakeholders considered that the participants engaged in deep reflections about the concept of universities and came up with interesting ideas, both in the Evocative and Re-Imagine activities. The results also revealed the powerful role of embodiment, not only by promoting unexpected connections but also in the general experience of the Re-Imaginatorium.

  One example from the evocative moment was the The Dynamic, Temporal, Rhythmical (musical) university. This university pays close attention to the flow, rhythm, and dynamic unfolding of teaching activities. A dynamic similar to music is used as a guide for how to sequence learning. For example, after intense periods (exams or intense lectures) students and teachers enter into a calm phase.



Figure 30. Example of composition from the Re-Imagine activity.



Figure 31. Example of composition from the Re-Imagine activity.



Figure 32. Example of composition from the Re-Imagine activity.

6. MAIN RESULTS AND FINAL DESIGN CONCEPT

6.1 Main results from the design process

The main results of the design process can be categorized into three, some are related to the format, some to the materials, and others to the collaboration with the stakeholders.

- The format: The three moments hold significant importance when it comes to enabling Critical Imagination. Each one contributes uniquely, and together they work in synergy. The first moment of contextualization laid a broad foundation of what could be imagined. The radical and in some cases, fun examples expressed that everything could be possible allowing participants to not limit their imagination.

  The evocative moment opened up and warmed up participants' imaginations. It allowed them to dream big or think about unreal ideas without reality checks. The unexpected combination of materials and the information in them enabled participants to find connections that usually are not seen.

  Finally, the Re-imagine moment helped to slowly land into more concrete thoughts and reflections about current practices and how they could be different in the future.

- The materials: The concept of ambiguity was demonstrated to be a potential tool to design materials that spark the imagination. The three types of ambiguity concerning the Critical Design Practices also proved to be useful at the moment to create thought-provoking designs. These two in combination can promote Critical Imagination.

  Furthermore, the physicality of the materials required specific qualities so they could support not only each of the moments of the format but also the possibility of replicating the Re-imaginatorium with different themes, facilitators, participants, and settings. Accordingly, characteristics like modularity, accessibility, and customization were possible to achieve with the use of diverse types of paper as the main material and paper circuits as the principal technology used.

- The collaboration with stakeholders: Working with real stakeholders with a participatory approach brought both, opportunities and challenges. The design concept benefited by having first-hand information from real users and other stakeholders. Furthermore, the overall process was not only a journey for developing a product but a learning opportunity for all parties involved.

6.2 Design Concept

Concept overview: The final design concept proposes a toolkit that guides and supports the creation of playful materials that aid the different moments of the format of the Re-Imaginatorium.

It was developed in collaboration with a team of stakeholders and is still in its development phase.

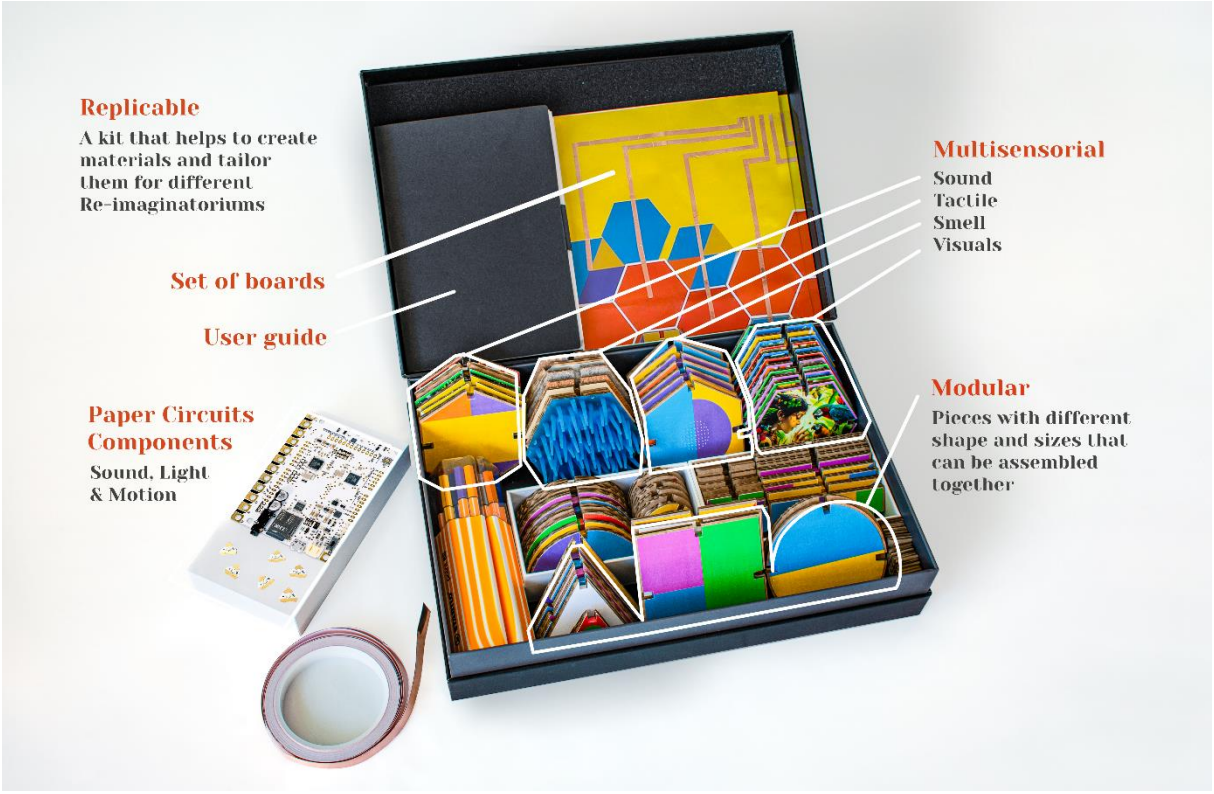


Figure 33. Sketch of Concept Design from the third prototype

Details

Thought-provoking and open interpretation:

The different elements in the materials are created to spark either critical thinking, imagination, reflection, or a combination of them. This is by employing ambiguity as a tool to design the tangible features. Ambiguity enables multiple interpretations that in turn promote appropriation, one relevant characteristic of a playful attitude.

- Playful approach: Appropriation is enhanced by ambiguous characteristics. These also promote experiencing a sense of exploration, discovery, and fantasy. This last one is bringing the impossible as possible.

Multisensorial:

This builds on the Thought-provoking characteristic. The materials support the use of different senses to enhance the thinking and reflective process from an embodied interaction. Visuals, textures, sounds, and smells evoke diverse sensations to be related to the context and theme of each Re-Imaginatorium.

- Playful approach: Sensory experience is a recognized way to bring meaningful interactions through the different senses.

Modular:

The main structure and system of the materials are modular so participants can create their own configurations. The pieces are to be assembled in two and three dimensions.

- Playful approach: Building upon the characteristic of appropriation. Modularity brings another layer to it by allowing participants to take abstract shapes, add meanings to them, and assemble them in ways that make sense to represent their ideas and thoughts, as well as show the relations among them.

  This also promotes experiencing a sense of creating something or expressing themselves creatively.

Replicable:

The concept of being a toolkit is precisely to gather different tools and provide guidance to support others in the creation of materials for Critical Imagination within the context and format of the Re-Imaginatorium. Visuals, words, smells, textures, sounds, etc. are to be tailored to the theme others might Re-Imagine.

- Playful approach: This characteristic supports the creative experience, not only in participants but also in facilitators who are the ones who would tailor the materials in diverse ways. This is a tool kit that starts by guiding how to develop those materials but ultimately is the facilitator and creator who after would have the ability and freedom to appropriate the kit and make creations that fulfill their purposes and ways of working.

The toolkit contains pieces that allow the change of visuals, textures, smells, sounds, and other possible features like light and motion. Sound, lights, and motion are to be created with the paper circuit elements within the toolkit.

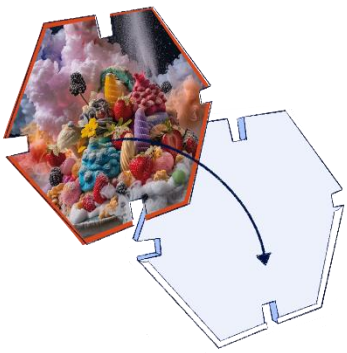


Figure 34. Pieces that can be tailored with diverse visuals and textures.

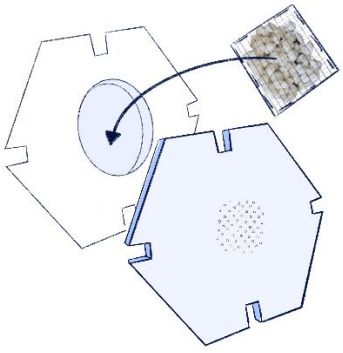


Figure 35. Pieces that can be tailored with diverse smells.

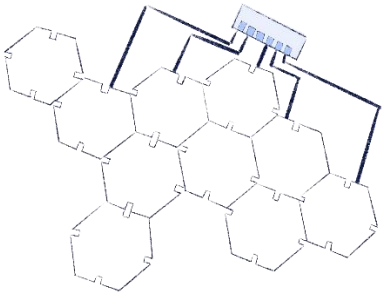


Figure 36. Tools that support the creation of materials with sounds, light, and motion.



Figure 37. Sketch of in-use materials for the Evocative Moment



Figure 38. Sketch of in-use materials for the Re-Imagine Moment

6.3 Answers to Research Questions

1. What role could Playful Design have concerning speculative thinking in a participatory setting?

Based on the research and empirical results, Playful Design has been shown to have the potential role as an enabler and enhancer of critical thinking and the speculation of different futures through its characteristic of appropriation. This helps to create playful materials and experiences that are thought-provoking and broaden the spectrum of what is possible to imagine. When incorporated in a participatory setting these materials act as a tangible common ground (Brandt, 2006) to explore, reflect, and build together possible meanings while interacting with them.

Additionally, playfulness can act as a bridge that allows the combination of speculative and participatory-oriented design approaches. This is in response to frequent criticism of the

speculative design of being elitist (Hillgren et al., 2020) and not involving the people impacted by the implications of the issue they address (Gerber, 2018, as cited in Hillgren et al., 2020). Through a playful approach not only designers are invited to speculate and imagine but it opens up the spectrum of who can participate.

Furthermore, motivation is a crucial aspect when setting up participation, a way potential way to promote it is via a playful experience which Lucero & Arrasvuori (2010) state to be per se highly motivating.

2. How Playful Design can make sense in the critical reflection and future creation of different aspects of human ways of living?

Besides the above, it can be stated that even when embracing challenging questions, approaching them with a playful attitude contributes to an enjoyable experience. Such experience supports the creation of a safe and flexible atmosphere, which is known to foster creative processes (Brandt, 2006). With it, the possibility to critically reflect on current human practices and meet unexpected visions of the future with a more open mindset.

Furthermore, this playful environment not only facilitated the generation of concrete ideas but also enabled participants to reflect on critical issues at a more abstract level. Engaging in such critical reflection holds already significant importance within the intention behind critical thinking.

3. What is the relation between playfulness and critical design practices?

Playfulness and the critical design practices have in common the creation of objects or systems that allow multiple interpretations, both considering ambiguity as a core resource to doing it.

Furthermore, the significance of playfulness in the critical design context lies in its ability to avoid forcing the activities into the category of mere criticality. Instead, it encourages participants to be open to unexpected combinations and unconventional ideas that capture their interest. Then, this openness allows them to critically reflect on why these ideas are interesting and which might be viable possibilities worth implementing or exploring further.

The ideas and reflections that emerged thanks to the openness that playfulness brought, have criticality at their core. For example, why is a night radio university relevant? This idea is critical by suggesting listening rather than the dominant sense of sight as the foundation for knowledge production. Or what is behind the idea of a self-awareness lobby? It highlights the often neglected aspect of mental health among students and university staff and criticizes the conception of being always in a productive state.

Finally, criticality usually comes from bringing in alternative perspectives to the dominant ones. This is not an easy task as we have internalized some perspectives and it takes time to experience what others' perspectives could be like. However, there was the sense in the workshops that the playful aspects, for example randomness, quickly brought unexpected perspectives allowing participants in a way experience them while critically making sense of the materials given.

6.4 Knowledge contribution:

This thesis project contributes to the field of playful Design on three levels. First, by bringing insights on the possibilities of combining Playful Design attributes, Critical Design Practices purposes, and Participatory Design approaches, in a way that makes sense when promoting critical thinking and imagination. Particularly, how the concepts of appropriation and ambiguity can be useful resources.

Secondly, by presenting insights from practical experience on how to plan and use methods to design playful materials for enhancing critical thinking and imagination.

Thirdly, by developing a method that can continue being utilized by the Re-Imaginatorium or for others when looking to spark critical thinking and imagination.

7. EVALUATION/DISCUSSION

Final outcome:

Overall the materials and the moments proposed for the format brought strong insight about what is needed to support Critical Imagination within the context of the Re-Imaginatorium. The concepts of appropriation and ambiguity proved to be possible design resources to enhance imagination. To my knowledge, they have been used within the design field however not in the combination and approach used in this thesis project. Furthermore, from the prototyping and user testing the playful materials and atmosphere in the workshops were highly appreciated by participants. This considerably enhanced the outcomes of the imagined futures.

However, different aspects should be reconsidered or further evaluated to fully understand the impact of the interaction with the materials, the moments of the format, and the playful characteristics applied to them.

When it comes to engagement, playful experiences are known to be per se motivating (Lucero & Arrasvuori, 2010), however, a reflection is worth doing here. Even when the testing was done with real users, they represented only a part of the broad intended users of the Re-Imaginatorium. Students and teachers from diverse programs and universities were the main participants in the three workshops. Most of them were between the ages of 20 to 25 and had certain experience in workshopping or other ideation activities. Consequently, it would be needed to test how the materials, the format, and their playful and embodied approach work with different segments of users, like regular citizens, workers in the private or public sector, the elderly, etc., who may not be familiar with these type of methods and practices.

Another reflection comes from the tangibility and affordance of materials. For the evocative moment, they successfully worked as intended, however, for the re-imagine moment they took another role. On one hand, it was a positive sign as the materials proved to be easy to appropriate. However, on the other hand, they did not fully support the visualization aspect of the scenario creation. Participants re-imagined universities, created concrete future ideas, and materialized them into tangible compositions that showed the different elements and their relations, but these compositions were not figurative enough to stand alone without requiring a verbal explanation.

Exploration, creation, and sensation were some of the identified aspects from user testing to contribute to a playful experience. The last one which involved different senses definitely took a central role in the third prototype and the final design concept. This supported also the playful experiences with a more embodied approach. However, Embodied Design is a broad field that requires in-depth study to adequately be implemented, which due to time constraints

was not conducted. Furthermore, due to the relevant results, research deeper into it is a clear step forward in the development of the Re-Imaginatorium.

Finally, some of the interactivity elements of the final concept were not tested. The craft of multiple materials was time-consuming and the lack of technical resources in the time frame for the scheduled workshops led to the prioritization of prototyping only some aspects. Consequently, further iterations are needed to fully test the complete set of the proposed elements in the concept.

Design process:

Overall, the iterative nature of the process supported the purpose and schedule of stakeholders. However, it turned into a complex project that required more time to fully develop. The design plan guided reasonably the process, and by conducting the selected methods, relevant insights were discovered, synthesized, and materialized into coherent design decisions. Some of the methods were adapted so they could contribute better to the process, context, and personal style.

There are anyhow, relevant aspects that need improvement. One is concerning time management and prioritization. This thesis project was influenced by the needs and schedules of stakeholders offering valuable opportunities to test concepts and materials with real users in authentic Re-Imaginatorium setups. However, this also increased the project's complexity, requiring additional time for comprehensive development. Due to time constraints, some activities were compromised. For instance, more thorough experimentation with paper circuits and other technologies would have been beneficial for the materials created for the prototypes.

Furthermore, there was a lack of tools for fully evaluating and analyzing the results of the user testing. Observations and feedback from participants brought valuable insights as well as confirmed or challenged some of the assumptions. However, more specialized methods would have been useful at the moment of testing, for example, the levels of engagement and the effect of certain interactions with the materials.

Stakeholders' collaboration:

The weekly brainstorming sessions with the main team of stakeholders and the workshops with real users were proved to facilitate the design process in a participatory way. The whole collaboration was perceived by the stakeholders as smooth and productive. The stakeholders involved gained relevant insights and learned along the process. This is a core aspect of a participatory design approach. As well as other aspects to improve in this thesis project, the collaboration would have been improved with more time, for example, by allowing to have more workshops with other relevant stakeholders or by expanding the spectrum of stakeholders involved in the whole process.

8. CONCLUSION

This thesis project aimed to support the development of the format and the materials for the Re-Imaginatorium through Playful Design. The Re-Imaginatorium is an initiative led by Malmö University as part of the UNIC alliance and has the purpose of opening collaborative spaces for Critical Imagination. In a series of workshops, the Re-imaginatorium invites different actors from academia, the public and private sectors, and citizens to Re-imagine how diverse aspects of our societies could be otherwise. Usually, the ideas that come up from these sessions are portrayed in scenarios and start with a "what if" question.

The first pilot of the Re-Imaginatorium inspired this thesis project. This pilot revealed an opportunity arising from the constraints encountered during the process of imagining and ideating how things could be different. In parallel, playfulness was perceived as a potential way to approach this challenge. Consequently, the main purpose of this thesis project was to research what role could Playful Design have concerning speculative thinking in a participatory setting.

The process was conducted collaboratively with a team of main stakeholders. Weekly brainstorming and sync sessions, complemented by three workshops with real users, were some of the main methods of collaboration. Exploration and iteration were relevant aspects of the process. Therefore, three prototypes and user testing were implemented, each one building up from the previous one.

From the theoretical and contextual analysis, one of the main milestones was the identification of ambiguity as a resource that bridges playful design and the design critical practices, through the play affordance of appropriation. This in turn was recognized as an enabler of imagination and critical reflection. Subsequently, another relevant milestone was the translation of these notions into tangible and coherent aspects for the format and materials for the Re-imaginatorium. This was done through a combination of moments within the format (one for contextualizing, another for evocating, and a last one for Re-imagining), the activities for them, and the supporting materials. These playful materials were open to interpretation, revealed unexpected connections, and supported embodied exploration.

From the results of the workshops with real users, it can be stated that the playful approach in the format and the materials, enabled and enhanced imagination and critical reflection among participants. This was observed in the quality of results and discussions.

The Re-imaginatorium is still in its development phase, however, valuable insights were generated with these experiences. For example, playfulness was considered by stakeholders as a significant and essential notion that opened new possibilities in the further conceptualization of the Re-imaginatorium. Furthermore, embodiment was recognized as a powerful approach at the moment of imagining and reflecting in participatory settings. This last one was not fully explored in the time frame of this thesis and needs further development.

ACKNOWLEDGEMENTS

I would like to express special gratitude to my supervisor Per-Anders Hillgren for not only his constructive suggestions during this study but also motivated me to have a playful approach myself toward the process and its results. I would also like to thank the team of stakeholders that allowed me to be part of the Re-Imaginatorium and let me explore with them its possibilities. Special thanks to all participants of the workshops whose valuable feedback and open attitude were an essential aspect of the process and concept.

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APPENDIX 1

1. Taxonomy of Ambiguity by Gaver et al. (2003)

Source: Gaver, W. W., Beaver, J., & Benford, S. (2003). Ambiguity as a resource for design.

TACTICS
Ambiguity of Information • Use imprecise representations to emphasise uncertainty • Over-interpret data to encourage speculation. • Expose inconsistencies to create a space of interpretation • Cast doubt on sources to provoke independent assessment
Ambiguity of Context • Implicate incompatible contexts to disrupt preconceptions • Add incongruous functions to breach existing genres • Block expected functionality to comment on familiar products
Ambiguity of Relationship • Offer unaccustomed roles to encourage imagination • Point out things without explaining why • Introduce disturbing side effects to question responsibility

2. Taxonomic matrix for critical design practice by Malpass (2013)

Source: Malpass, M. (2013). Between Wit and Reason: Defining Associative, Speculative, and Critical Design in Practice.

Table 1 Taxonomic matrix for critical design practice.

Practice	*Method: definition*	*Type of satire: satiric mechanism*	*Type of ambiguity*	*Object rationality*
Associative design	Cut-up: When one or more objects are cut up or reassembled to exaggerate their properties and give new meaning Context transfer: When one object is taken out of context and placed into another	Horatian: Parody Burlesque *Reductio ad absurdum*	Ambiguity of context	Rational Familiar archetype
Speculative design	Hybridity: One archetype integrated with another archetype. This might take the form of two objects but also practices. For example, technology that exists in a laboratory context is placed in a quotidian setting. Technocratic visualization: Technocracy is a wide-ranging visual system that is legitimized by specific reference to scientific expertise. The science rationalizes the proposition.	Horatian into Juvenalian: Narrative Distortion Exaggeration Allegory Anticlimax	Ambiguity of information	
Critical design	Extrinsic narrative: A narrative is established to situate the object. Questions are raised in the difference between "reality" and the materiality proposed through the object and its narrative of use.	Juvenalian: Allegory Violence Obscenity Antithesis Prolepsis	Relational ambiguity	Nonrational Unfamiliar archetype

Note: For definitions and discussions on the type of satire, see Simpson (2003) and Connery and Combe (1996).

3. PLEX Framework: Characteristics of a Playful Experience by Korhonen et al. (2009)

Source: Korhonen, H., Montola, M., & Arrasvuori, J. (2009). UNDERSTANDING PLAYFUL USER EXPERIENCE THROUGH DIGITAL GAMES.

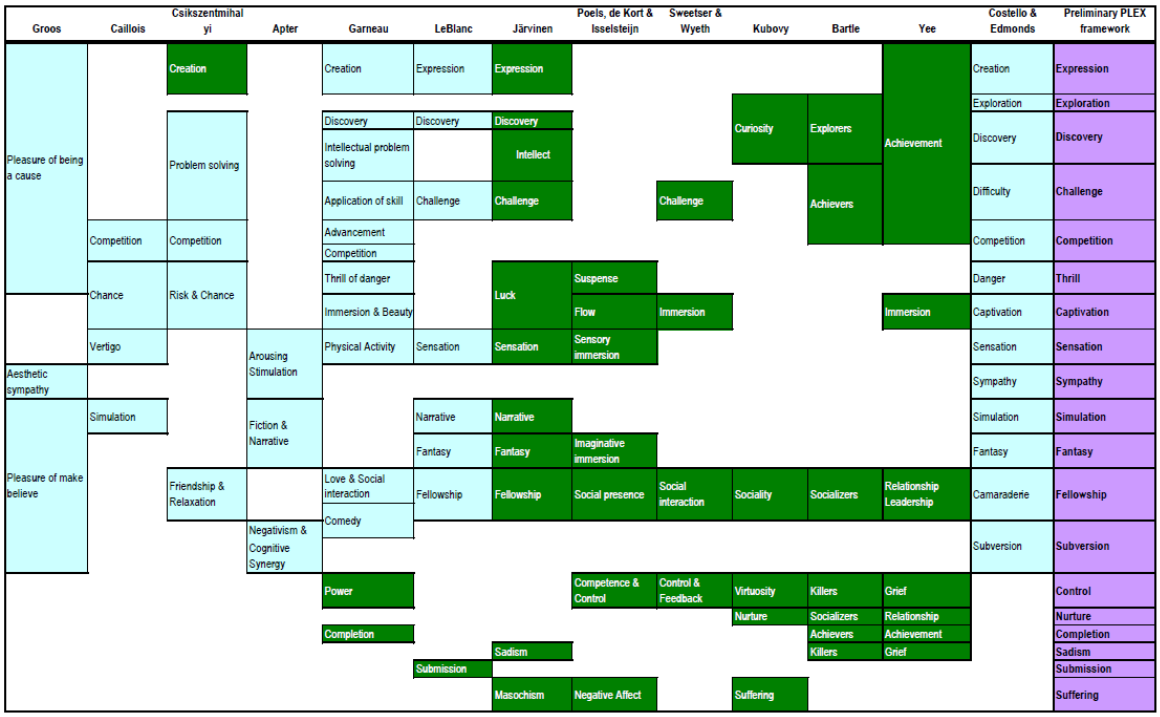


Figure 1 Comparative analysis used as a foundation of the study described in this paper. Turquoise boxes describe the Costello and Edmonds's model. Green boxes are additions we have included to the analysis of playful experiences. See Table 2 for definitions of each playful experience category

Table 2 Definitions of PLEX Framework Categories

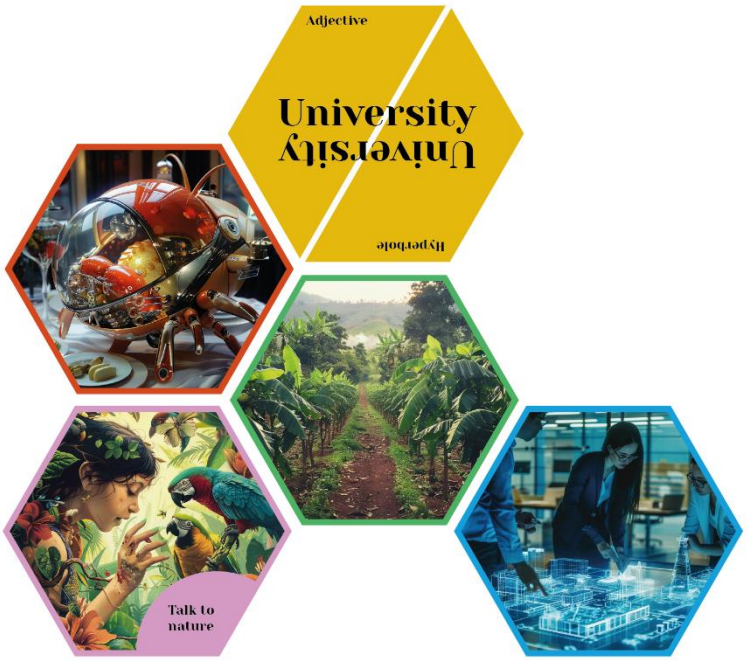
Category	Description
Captivation	Experience of forgetting one's surroundings
Challenge	Experience of having to develop and exercise skills in a challenging situation
Competition	Experience of victory-oriented competition against oneself, opponent or system
Completion	Experience of completion, finishing and closure, in relation to an earlier task or tension
Control	Experience power, mastery, control or virtuosity
Discovery	Experience of discovering a new solution, place or property
Eroticism	Experience of sexual pleasure or arousal
Exploration	Experience of exploring or investigating a world, affordance, puzzle or situation
Expression	Experience of creating something or expressing oneself in a creative fashion
Fantasy	Experience of make-believe involving fantastical narratives, worlds or characters
Fellowship	Experience of friendship, fellowship, communality or intimacy
Nurture	Experience of nurturing, grooming or caretaking
Relaxation	Experience of unwinding, relaxation or stress relief. Calmness during play
Sadism	Experience of destruction and exerting power over others
Sensation	Meaningful sensory experience
Simulation	Experience of perceiving a representation of everyday life
Subversion	Experience of breaking social roles, rules and norms
Suffering	Experience of frustration, anger, boredom and disappointment typical to playing
Sympathy	Experience of sharing emotional feelings
Thrill	Experience of thrill derived from an actual or perceived danger or risk

APPENDIX 2

The visuals for six sets of cards used in the different workshops.






Adjective
University
Hyperbole
Talk to nature


Advance sense of smell
Adjective
University
Hyperbole


Adjective
University
Hyperbole
Mix
reality


Adjective
University
Hyperbole
Regenerative
thinking